

Final Submission – Cloud Computing

Cloud Foundations
Noroff

by
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Table of Contents

1 Introduction.....	3
2 Main Part.....	3
2.1 Core Cloud Infrastructure Design.....	3
2.1.1 Design Principles.....	4
2.1.2 DNS Configuration.....	5
2.1.3 Virtual Network Infrastructure.....	5
2.2 Identity and Access (IAM).....	8
2.2.1 Hybrid Identity Model.....	8
2.2.2 Conditional Access and RBAC.....	10
2.3 Security, Availability and Operations.....	11
2.3.1 Security Requirements.....	11
2.3.2 Availability Strategy.....	12
2.3.3 Monitoring and Logging.....	12
2.4 Department-Based Cloud Services.....	13
2.4.1 Internal Web Application.....	13
2.4.2 Secure Document Sharing.....	15
2.4.3 Isolated Dev/Test Environment.....	19
2.4.4 Public Support Chat.....	20
2.4.5 Internal AI Assistant.....	22
2.5 Data and Analytics.....	24
2.5.1 Data Lake Architecture (Blob Storage).....	24
2.5.2 Analytics with Fabric.....	25
2.5.3 AI and Data.....	26
2.6 Use of AI.....	27
2.7 Cloud Journey Reflection.....	27
3 Summary and Conclusions.....	28
4 References.....	30

1 Introduction

This report presents a cloud and hybrid target architecture for NerdyClouds. The goal is to improve daily operations, security, and scalability through selective use of Azure services, supported by an identity and connectivity foundation.

It outlines the core cloud design and the principles guiding key decisions. It describes the networking approach, including DNS design and virtual network structure, to ensure predictable connectivity between services and environments. It then describes the identity and access model, where Microsoft Entra ID (Entra ID), group-based access, and role-based controls provide a way to manage authentication and authorisation (Microsoft, 2024a; Sandhu, 1997).

Security, availability, and operational controls are designed around least-privilege access, managed services, and monitoring to help meet the uptime target and reduce day-to-day maintenance (NIST, 2020a). The report also covers key department-facing services, including an internal web application, secure document sharing using Azure Storage and Blob containers, and an isolated environment for development and testing.

To support data-driven decision-making, the solution includes a blob-based data foundation and an analytics demonstration using Fabric and Power BI. The solution also introduces two AI-enabled services: a public support chat using Azure AI Language (Question Answering) and a restricted internal AI assistant hosted in Microsoft Foundry.

Proof-of-Concept implementations and validation steps are included throughout to show that access control and service behaviour work as intended.

2 Main Part

This section details the target architecture and the decisions behind identity, networking, and service design. Where features require licensing or enterprise connectivity, they are described as target design and validated conceptually.

2.1 Core Cloud Infrastructure Design

The core infrastructure design is based on NerdyClouds' current size and operational needs, balancing security, scalability, and simplicity while supporting future growth. The environment uses repeatable patterns for resource groups, identity, and networking to keep delivery consistent and manageable. This structure also supports infrastructure as code, which is demonstrated in the next section.

2.1.1 Design Principles

The infrastructure is designed around identity-based access control, a central connectivity model, and clear separation between environments to support secure access for both on-site and remote users (NIST, 2020b).

Several network models were evaluated before selecting hub-and-spoke. The choice was based on scalability needs, operational simplicity, and the requirement for centralised control. The hub provides shared connectivity, while separate VNets keep environments separated and simplify traffic control (Microsoft, n.d.-b).

Infrastructure as Code supports repeatable deployments, so environments can be recreated consistently and teams work from the same baseline (Artac et al., 2017). To demonstrate, the core environment was deployed using an ARM orchestrator executed via Azure CLI. The orchestrator retrieves stored templates from a dedicated Azure Storage account and runs them as one subscription-level deployment, demonstrating consistent deployment and reducing manual configuration differences (Mumbere et al., 2024; Microsoft, 2025a).

Figure 1 ARM orchestrator executed via Azure CLI, deploying the core environment successfully.

```
erik [ ~ ]$ az deployment sub create \
  --name nc-orchestrator \
  --location norwayeast \
  --template-uri "https://nctemplatesiac.blob.core.windows.net/templates/NC-IaC.json?sp=r&st=2026-02-06T10:09:41Z&se=2026-02-06T18:24:41Z&spr=https&sv=2024-11-04&sr=b&sig=2Gkl%2Bc6ggAhntoAKvWEQMOZ%2BRm%2BeHL3tBQBqFOSdEbY%3D"
{
  "id": "/subscriptions/8139006c-b185-486a-9d23-5526a99a7cf9/providers/Microsoft.Resources/deployments/nc-orchestrator",
  "location": "norwayeast",
  "name": "nc-orchestrator",
  "properties": {
    "correlationId": "51d00d93-e528-4a9a-a55f-0b4cf8eed347",
    "debugSetting": null,
    "dependencies": [
      {
        "dependsOn": [
        ],
      },
    ],
    "provisioningState": "Succeeded",
    "templateHash": "13806193286820751671",
    "templateLink": {
      "contentVersion": "1.0.0.0",
      "id": null,
      "queryString": null,
      "relativePath": null,
      "uri": "https://nctemplatesiac.blob.core.windows.net/templates/NC-IaC.json?sp=r&st=2026-02-06T10:09:41Z&se=2026-02-06T18:24:41Z&spr=https&sv=2024-11-04&sr=b&sig=2Gkl%2Bc6ggAhntoAKvWEQMOZ%2BRm%2BeHL3tBQBqFOSdEbY%3D"
    },
    "timestamp": "2026-02-06T10:27:38.892729+00:00",
    "validatedResources": null,
    "validationLevel": null
  },
  "tags": null,
  "type": "Microsoft.Resources/deployments"
}
```

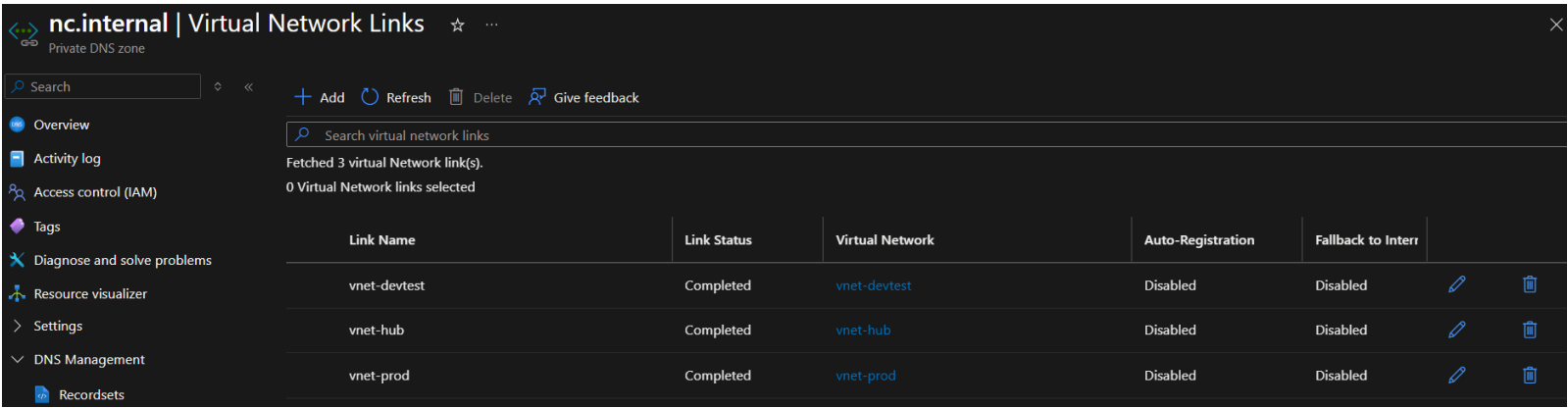
2.1.2 DNS Configuration

DNS is a foundational component in the proposed hybrid architecture, enabling name resolution across on-premises and cloud environments. Reliable DNS resolution is essential for service uptime and communication between distributed services (NIST, 2020a).

Azure Private DNS is used as the primary cloud DNS solution due to its cloud-native design and built-in fault tolerance. As a platform-managed service, Private DNS is intended to help avoid customer-managed DNS servers and reduce common single points of failure. Private DNS zones are linked to the relevant virtual networks in the environment to ensure internal service names are resolved privately without exposing resources to the public internet (Microsoft, 2025c).

In the hybrid setup, on-premises Active Directory Domain Services DNS hosts the internal DNS zone data, and is used as the primary resolver for internal names (Microsoft, n.d.-c). In a production environment, this can be extended through conditional forwarding over the Site-to-Site VPN, where on-premises DNS forwards queries for Azure private namespaces to be resolved in Azure (Immidisetti, 2021; Microsoft, n.d.-c). This approach is intended to provide resilient and secure name resolution across the hybrid environment.

Figure 2 Private DNS zone linked to Hub/Prod/DevTest-VNets.

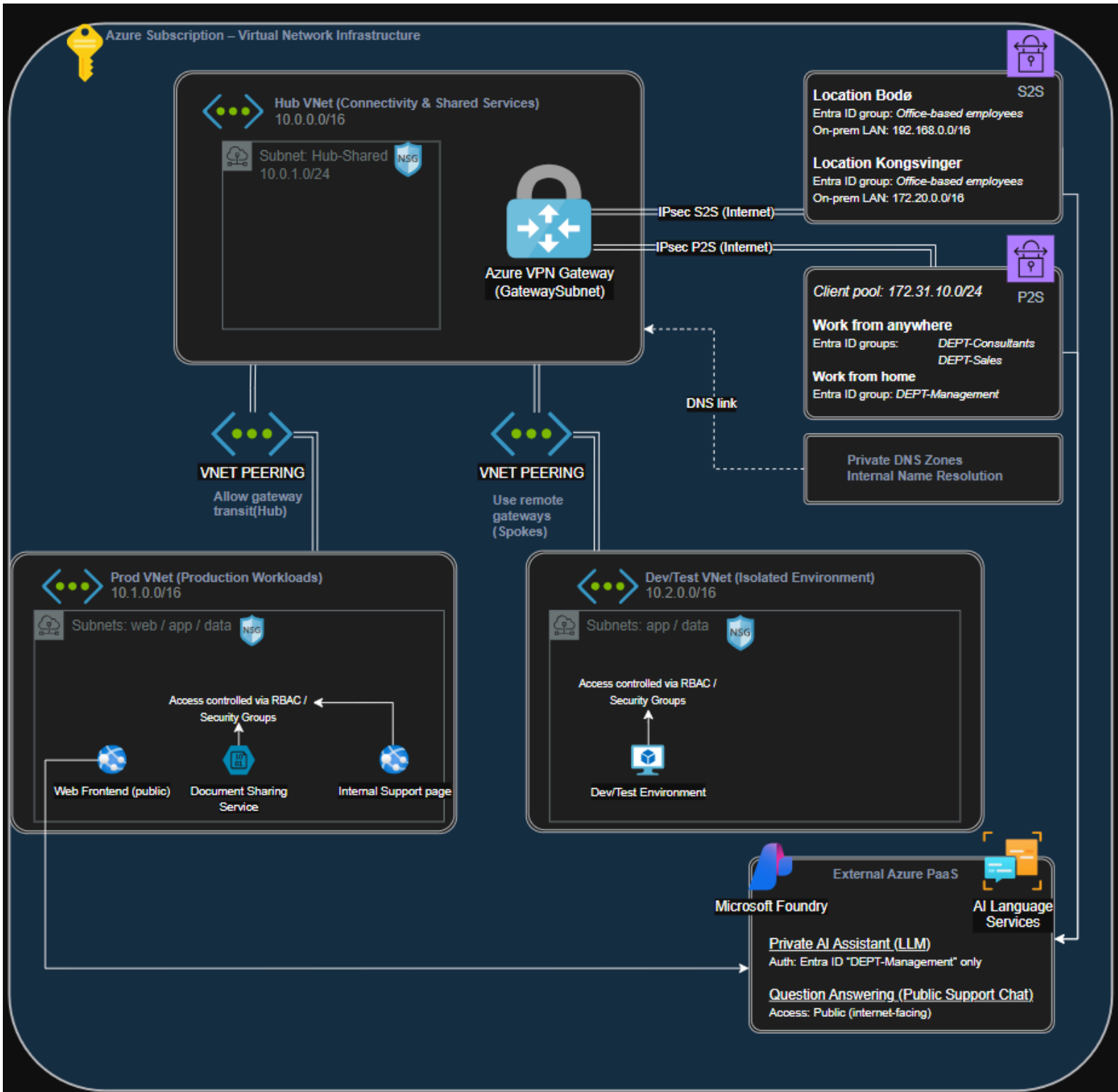


nc.internal Virtual Network Links						
Private DNS zone						
Search						
+ Add Refresh Delete Give feedback						
Overview						
Activity log						
Access control (IAM)						
Tags						
Diagnose and solve problems						
Resource visualizer						
Settings						
DNS Management						
Recordsets						
Search virtual network links						
Fetched 3 virtual Network link(s).						
0 Virtual Network links selected						
Link Name	Link Status	Virtual Network	Auto-Registration	Fallback to Intern		
vnet-devtest	Completed	vnet-devtest	Disabled	Disabled		
vnet-hub	Completed	vnet-hub	Disabled	Disabled		
vnet-prod	Completed	vnet-prod	Disabled	Disabled		

2.1.3 Virtual Network Infrastructure

The infrastructure is designed using a hub-and-spoke topology (Microsoft, n.d.-b). Figure 3 shows how connectivity, environment separation, and internal name resolution are implemented in the proposed hub-and-spoke network design, including the IP and subnet plan.

Figure 3 Azure Virtual Network infrastructure with hub-and-spoke topology.

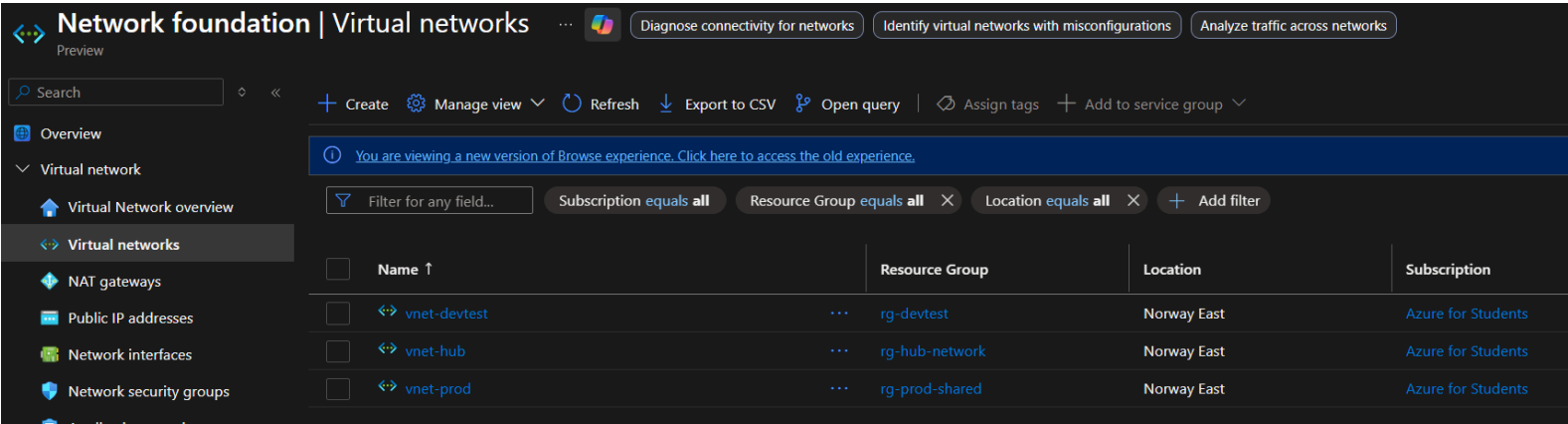


The Hub-VNet functions as the central hub for connectivity and hosts the Site-to-Site VPN gateway, serving as a central control point for inbound traffic and helping simplify security management and troubleshooting (Microsoft, n.d.-b).

Production services are placed in Prod-VNet to isolate operational services from other environments. The DevTest-VNet provides an isolated environment for development and testing, reducing the risk of non-production services from impacting live services (Microsoft, n.d.-b).

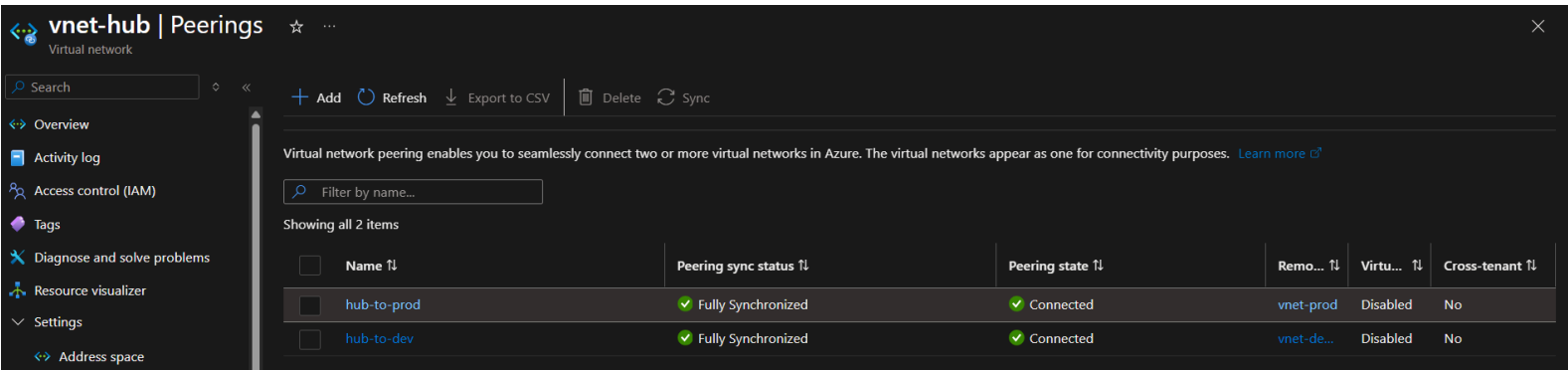
Connectivity between the Hub-VNet and the spoke VNets is established using VNet peering. Peering is limited to hub-to-spoke connections, helping maintain centralised control and logical isolation between environments (Salko, 2024; Microsoft, n.d.-b). As described in Section 2.1.2, the Private DNS zones are linked to the different VNets to enable private resolution across the segmented network architecture.

Figure 4 Hub-and-spoke VNets overview.



☐	Name ↑	Resource Group	Location	Subscription
☐	↔ vnet-devtest	rg-devtest	Norway East	Azure for Students
☐	↔ vnet-hub	rg-hub-network	Norway East	Azure for Students
☐	↔ vnet-prod	rg-prod-shared	Norway East	Azure for Students

Figure 5 Hub-to-spoke VNet peering.



☐	Name ↑	Peering sync status ↑	Peering state ↑	Remo... ↑	Virtu... ↑	Cross-tenant ↑
☐	hub-to-prod	✓ Fully Synchronized	✓ Connected	vnet-prod	Disabled	No
☐	hub-to-dev	✓ Fully Synchronized	✓ Connected	vnet-de...	Disabled	No

Hybrid connectivity is designed using a Site-to-Site VPN terminating in the Hub-VNet (Microsoft, n.d.-b; Microsoft, 2025o). This allows inbound traffic from on-premises environments to be routed through a single entry point, and traffic is distributed to the Production- and DevTest-VNets through VNet peering (Microsoft, n.d.-b). Within each VNet, traffic is segmented at the subnet level, and subnets are protected by Network Security Groups as a secondary

network control layer. Together with identity-based access controls, this supports least-privilege enforcement in line with Zero Trust principles (Microsoft, 2025d; NIST, 2020a; NIST, 2020b).

The internal AI assistant is hosted in Microsoft Foundry and accessed via secured API endpoints (Jackson, 2025). The public support chat uses Azure AI Language (Question Answering) and is exposed as a public endpoint (Microsoft, 2025e). This is shown conceptually in the network overview and is not part of the routed VNet path. Point-to-Site VPN is included in the target design to support secure remote connectivity to the Hub-VNet (Microsoft, 2025o). Access is enforced through identity-based controls in line with Zero Trust principles (NIST, 2020b).

The hub-and-spoke design reduces overcomplicated network structure and provides a control point for connectivity. For NerdyClouds, this helps to make changes and troubleshooting faster while keeping Production and DevTest separated. With connectivity and separation in place, access control is enforced through identity.

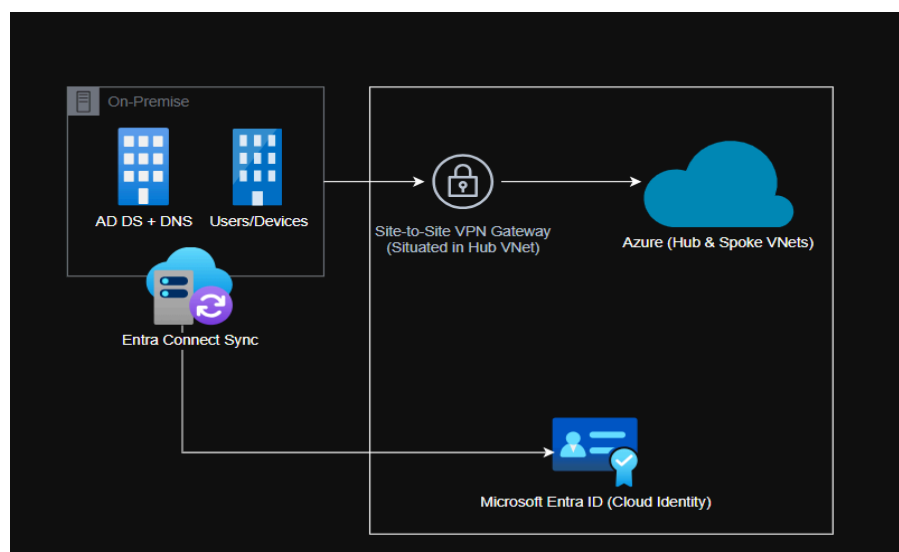
2.2 Identity and Access (IAM)

Identity and access are handled as one system, combining Entra ID, group-based access, Conditional Access, and role-based access control (RBAC) to enforce least-privilege access as roles change across the organisation (NIST, 2020a; Microsoft, 2024a; Beeram, 2025). Entra ID acts as the central identity platform, integrated with on-premises Active Directory through Entra Connect, and is intended to provide consistent authentication and access enforcement across hybrid and cloud-native services (Microsoft, 2025f).

2.2.1 Hybrid Identity Model

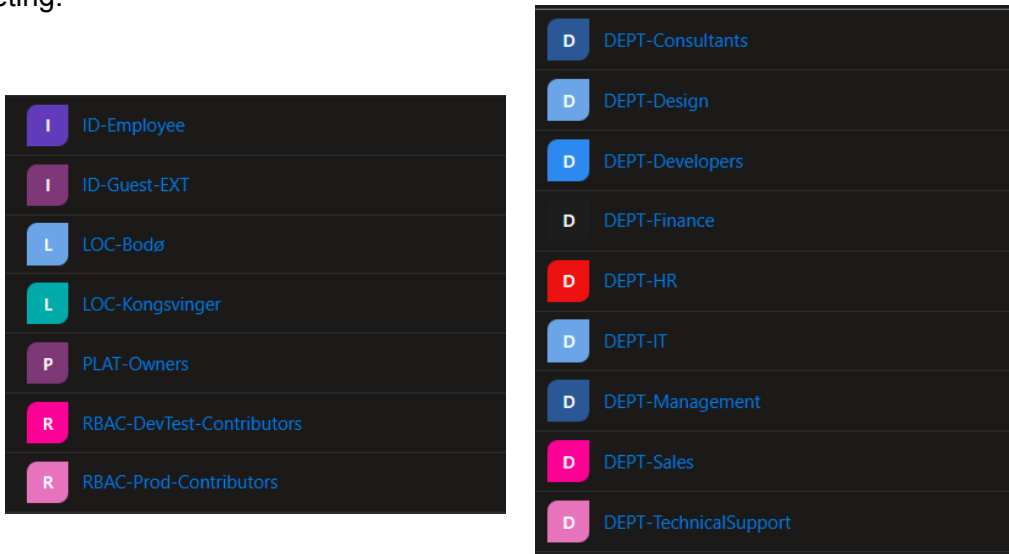
All employee accounts are created and managed on-premises and synchronised to Entra ID using Entra Connect. In this model, user accounts and security groups are synchronised to ensure that identity lifecycle management and group-based access control remain consistent across both environments. Entra ID acts as the central identity platform for authentication and access decisions, while group-based identity management is used throughout the environment to support predictable access control and auditing (Microsoft, 2025f, 2025g).

Figure 6 Hybrid identity and connectivity model.



As illustrated in Figure 6, on-premises Active Directory Domain Services remains the primary identity source, while Entra Connect synchronises identities to Entra ID for cloud authentication. A Site-to-Site VPN Gateway in the Hub VNet provides secure connectivity between on-premises and Azure services in the hub-and-spoke network (Borra, 2024; Microsoft, n.d.-b; Microsoft, 2025o).

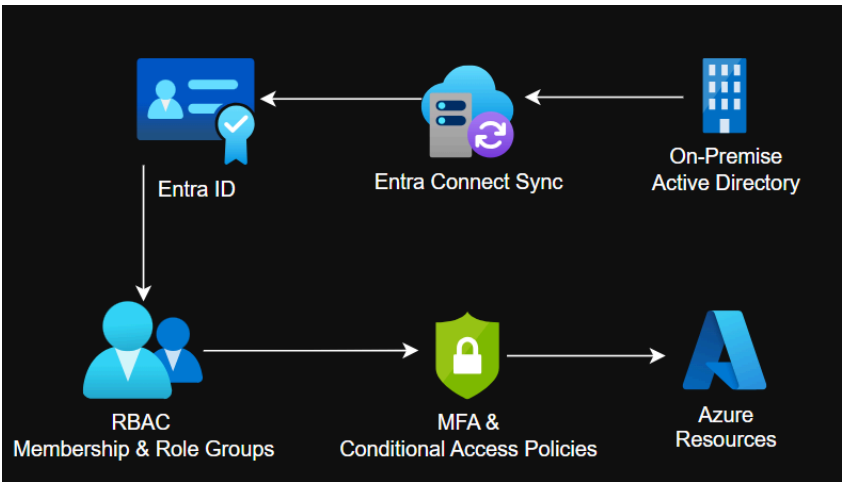
Figures 7 and 8 Entra ID security groups used for RBAC and Conditional Access targeting.



Conditional Access is used to enforce identity-based access rules, such as requiring multi-factor authentication and restricting insecure or non-compliant sign-ins (Howard et al., 2023). In this report, Conditional Access policies are described as target design and are not implemented as part of the Proof-of-Concept, as they add additional licensing cost and are not shown explicitly in the current student environment.

Least privilege is enforced by separating membership groups (such as department and location groups) from role-based groups used for RBAC (Microsoft, 2025b). Permissions are assigned to role groups rather than individual users, and access is granted by adding users to the appropriate groups. This reduces the risk of over-privileged accounts (NIST, 2020a). Access changes are handled through group membership updates instead of direct permission changes.

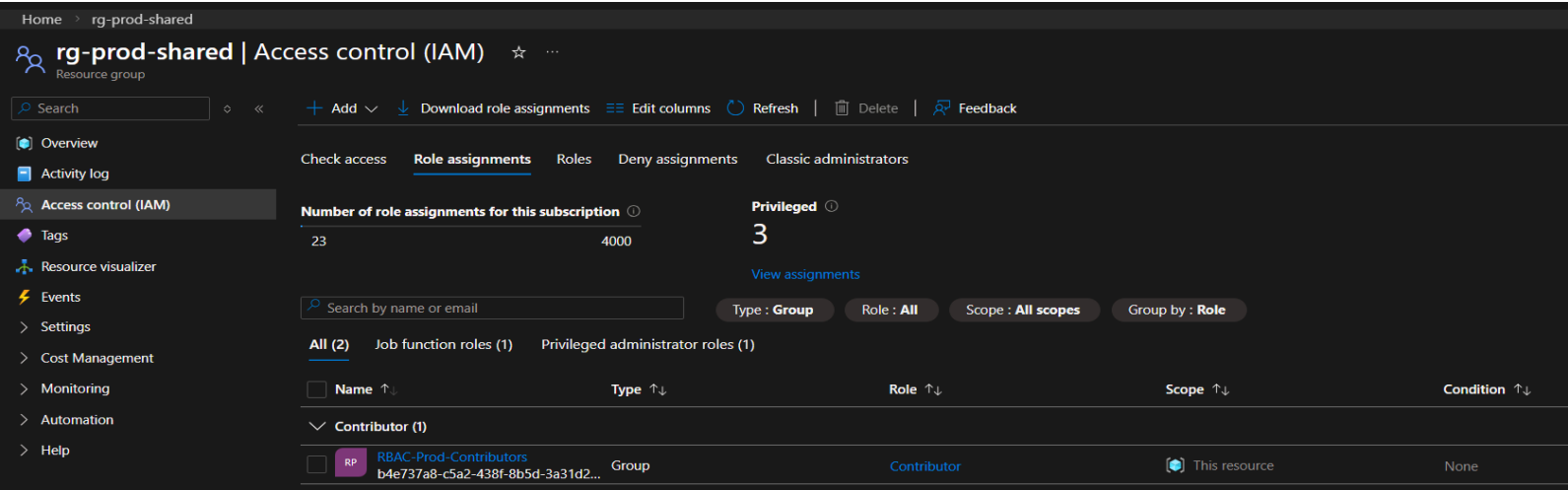
Figure 9 Identity flow and access management-diagram.



As shown in Figure 9, on-premises users and groups are synchronised to Entra ID using Entra Connect, and how group-based access together with Conditional Access policies is intended to be used to control access to cloud services hosted in Azure.

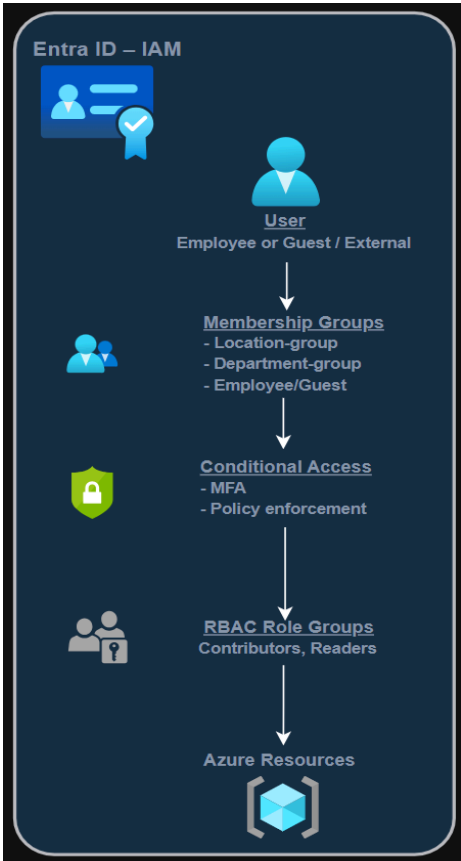
In this design, all employee accounts are created and managed in Active Directory as long as the organisation relies on on-premises user and group management. This helps to keep identities consistent across both on-premises and cloud environments and help prevent users from being created directly in Entra ID without an on-premises record. Using a single source for identity management is intended to make access control easier to manage, improve traceability for changes, and reduce the chance of inconsistent accounts in a hybrid setup.

Figure 10 RBAC role assignment applied to a group at resource group scope, enabling group-based access control.



2.2.2 Conditional Access and RBAC

Figure 11 IAM model showing how groups, Conditional Access, and RBAC work together to control access.



RBAC is applied through Entra ID security groups at subscription and resource group scope. Conditional Access enforces identity-based access rules, such as requiring multi-factor authentication and blocking legacy authentication protocols. Zero Trust is implemented through Conditional Access policies that evaluate identity, risk, and context before access is granted. (Microsoft, 2024a; Microsoft, 2025h; NIST, 2020b).

Remote access policies are aligned with department-based group structure. Users in DEPT-Management are permitted home-based access through Conditional Access policies combined with MFA enforcement. DEPT-Consultants and DEPT-Sales are permitted to authenticate from any geographic location, subject to stricter identity controls such as MFA and risk-based sign-in evaluation. Office-based departments such as DEPT-HR, DEPT-Finance, and Technical Support primarily access services through the on-premises network and the Site-to-Site VPN connection. In production, this can be enforced by Conditional Access policies that restrict authentication for office-based departments to trusted corporate locations (Microsoft, 2025h). Where internal-only Azure resources must be reached externally, Point-to-Site VPN provides secure remote connectivity for external users (Microsoft, 2025o). Resource access is enforced through Entra ID group membership and RBAC.

Remote access is achieved by authenticating users through Entra ID and enforcing Conditional Access policies, while network access is provided through the Hub-VNet connectivity model.

Overall, the IAM model supports consistent authentication and access control across both hybrid and cloud services. For NerdyClouds, this helps reduce manual access handling and make role changes easier to manage through group membership rather than individual permissions.

2.3 Security, Availability and Operations

Section 2.3 outlines the baseline security approach, availability strategy, and operational controls that support the target architecture. Identity and access controls are established in Section 2.2; this section focuses on how NerdyClouds maintains security, uptime, and operational visibility across the environment.

2.3.1 Security Requirements

In this design, security requirements are implemented through an identity-first approach using Entra ID (NIST, 2020b). Group-based authorisation and Conditional Access are defined in Section 2.2 and 2.2.2, and applied across the services described in Section 2.4. Authentication is required for internal services, while authorisation is limited to approved groups to enforce least privilege across departments (NIST, 2020a).

Security is treated as a shared responsibility between Microsoft and NerdyClouds, allowing the IT-team to focus on governance and access control rather than platform upkeep (Lane et al.,

2017). Managed services such as Private DNS, App Service, and Blob Storage are intended to reduce day-to-day maintenance by relying on Microsoft for service availability and platform maintenance. NerdyClouds remains responsible for configuration, identity governance, access assignments, and data handling.

From a compliance perspective, the architecture is designed to minimise exposure of sensitive data by default. Departmental document access is isolated using RBAC at container scope, and internal services are not exposed publicly unless required (Microsoft, 2024a). This solution helps to reduce risk by limiting both the attack surface and the number of identities with access to sensitive resources.

2.3.2 Availability Strategy

NerdyClouds requires a minimum uptime target of 98.5%. This is supported by using managed services in Azure that are intended to improve reliability and reduce maintenance. In Azure, Private DNS is used as the platform-managed DNS service for cloud name resolution (see Section 2.1.2). In the hybrid setup, on-premises AD DS/DNS remains authoritative for internal domains and is hosted on multiple domain controllers for redundancy.

For storage-based services, Zone-Redundant Storage is selected to improve resilience against zone-level failures while keeping the solution simple and cost-aware for a small organisation (Microsoft, 2025i). This is applied to the document-sharing storage account supporting Design and Management, intended to maintain availability even during localised infrastructure outages.

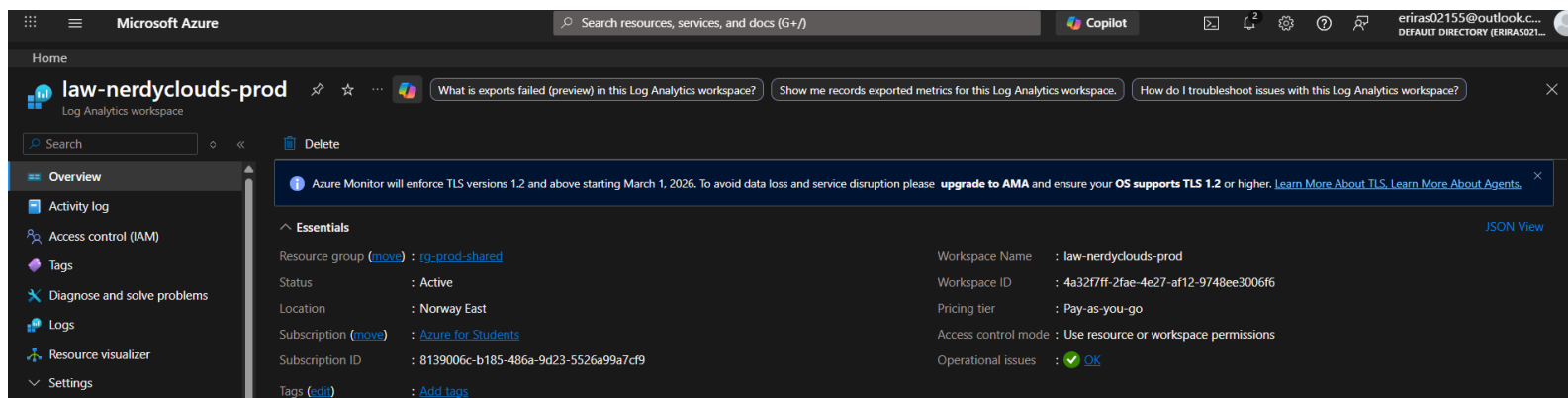
The DevTest-environment is intentionally designed with lower availability requirements than production, as its purpose is experimentation rather than business continuity. This results in resources being aligned with business criticality and helps avoid overengineering.

2.3.3 Monitoring and Logging

Effective security and operational control require continuous visibility into both identity-related activity and the state of the cloud environment (NIST, 2020a). Centralised monitoring and logging are intended to provide an overview of the environment, enabling insight into access events, configuration changes, and service health over time.

Azure Monitor and Log Analytics are used to collect and review monitoring data in this environment (Microsoft, 2025l). In this design, Azure-native monitoring helps minimise reliance on third-party tools and consolidate key events into a single workspace rather than across individual services. In this environment, diagnostic logging is enabled for key services such as App Services, storage accounts, and network components.

Figure 12 Log Analytics workspace providing a centralised view of logs and monitoring data.



Backup requirements are applied based on how important the service is. In this design, platform-managed services are treated as reducing backup complexity. Backup is therefore treated as an operational safeguard rather than a replacement for monitoring and visibility.

Together, these choices rely on managed services for platform availability, while Nerdyclouds keeps control of identity, access, and configuration.

2.4 Department-Based Cloud Services

Department services are implemented to support daily operations through secure access to internal tools, controlled document sharing, and clear separation between departments.

This section covers the internal web application, the document sharing solution using Azure Storage, and the two AI-enabled services used for support and management.

2.4.1 Internal Web Application

Technical Support needs an internal web application hosted in the cloud. The service must be available regardless of user location, while access is restricted to authorised internal users through Entra ID authentication and group-based authorisation.

The web application is hosted using Azure App Service (Web App). This managed platform was selected to help reduce day-to-day maintenance by removing the need for server administration, operating system patching, and platform upkeep, while providing built-in availability and scalability for an internal business-critical application. This supports the organisation's uptime target by relying on a Microsoft-managed service rather than a self-managed server (Microsoft, 2025j).

In this configuration, the app is set up to use Entra ID authentication. Before access is granted, all requests are authenticated using Entra ID, establishing an identity-based access boundary rather than relying on network location (Microsoft, 2025k). Authorisation is scoped through group-based assignment, limiting access to approved users only.

Figure 13 App Service (internal web app).

Home

techsupinternal Web App

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

Resource visualizer

Deployment

Settings

Environment variables

Configuration (preview)

Instances

Authentication

Identity

Backups

Custom domains

Certificates

Networking

WebJobs

Service Connector

Locks

Essentials

Resource group (move) : rg-prod-shared

Status : Running

Location (move) : Norway East

Subscription (move) : Azure for Students

Subscription ID : 8139006c-b185-486a-9d23-5526a99a7cf9

Tags (edit) : Add tags

Default domain : techsupinternal-gmhxaah3fwh7bydn.norwayeast-01.azurewebsites.net

App Service Plan : ASP-rgprodshared-ba10 (F1: 1)

Operating System : Linux

Health Check : Not Configured

Properties Monitoring Logs Capabilities Notifications (0) Recommendations

Web app

Name : techsupinternal

Publishing model : Code

Runtime Stack : Node - 24-lts

Runtime status : Starting

Domains

Default domain : techsupinternal-gmhxaah3fwh7bydn.norwayeast-01.azurewebsites.net

Custom domain : Add custom domain

Hosting

Plan Type : App Service plan

Name : ASP-rgprodshared-ba10

Deployment Center

Deployment logs : View logs

Last deployment : Loading deployments...

Deployment provider : None

Application Insights

Name : Not supported. Learn more

Networking

Virtual IP address : 20.100.2.128

Outbound IP addresses : 4.220.26.158, 131.163.59.213, 4.220.20.... Show More

Additional Outbound IP addresses : 4.220.26.158, 131.163.59.213, 4.220.20.... Show More

Virtual network integration : Not supported

Figure 14 App Service authentication (Entra ID).

techsupinternal | Authentication Web App

Search

Refresh

Troubleshoot

Send us your feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

Resource visualizer

Deployment

Settings

Environment variables

Configuration (preview)

Instances

Authentication

Identity

You can choose an identity provider to manage user identities and authentication flows. Add providers here, edit settings, and decide which provider is handling authentication for your app. Learn more

Authentication settings Edit

Requiring authentication ensures that requests to your app include information about the caller, but your app may still need to make additional authorization decisions to control access. If unauthenticated requests are allowed, any client can call the app and your code will need to handle both authentication and authorization. Learn more

App Service authentication : Enabled

Restrict access : Require authentication

Unauthenticated requests : Return HTTP 302 Found (Redirect to identity provider)

Redirect to : Microsoft

Token store : Enabled

Identity provider

Add provider

Identity provider : Microsoft (techsupinternal)

App (client) ID : f3cf0a38-5579-4d51-a8dc-24abde08d43c

Figure 15 Blocked access for an unauthorised user.

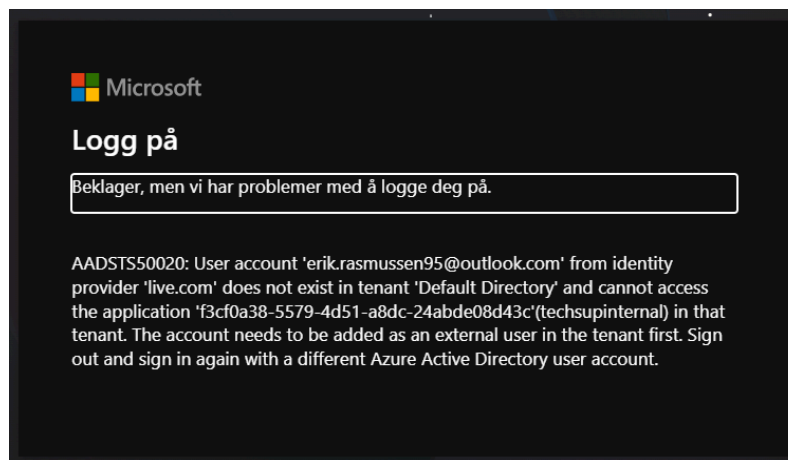
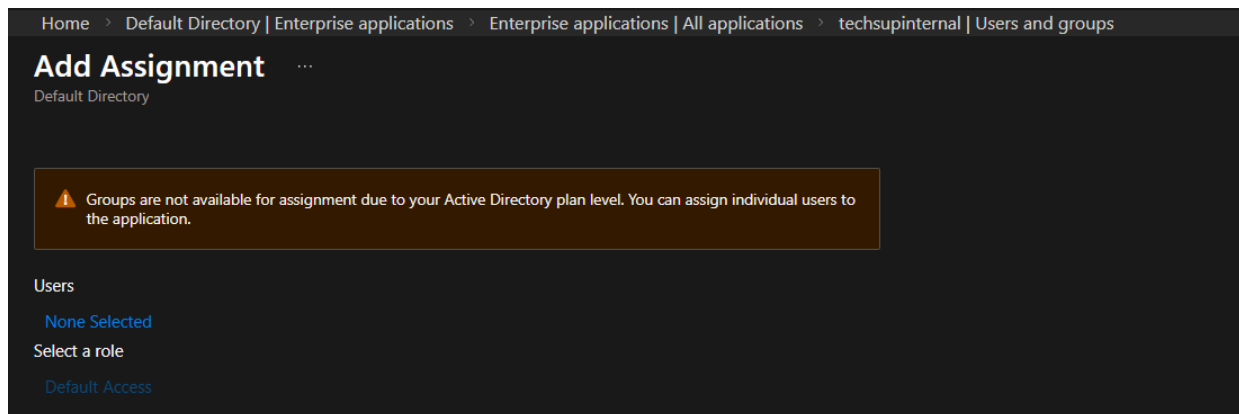


Figure 16 Tenant limitation for group assignment (Proof-of-Concept).



In production, access would be configured using “User assignment required” and approved groups (Microsoft, 2025k). To align with the hybrid DNS strategy detailed earlier, the App Service hostname could be mapped conceptually to an internal name such as *support.internal.nerdy.com*.

2.4.2 Secure Document Sharing

The Design and Management departments require secure document-sharing solutions with strict separation of access. Design needs to distribute reusable templates internally, while Management requires a private document workspace for files that must not be accessible to other departments.

Azure Blob Storage is selected as a simple, scalable storage foundation with Entra ID integration. A single storage account using Zone-Redundant Storage (ZRS) is deployed to support the organisation's 98.5% availability requirement (Microsoft, 2024b, 2025i). Departmental separation is implemented using separate blob-containers within the same storage-account.

Access is enforced using Entra ID group-based RBAC at container scope (Microsoft, 2024b). Users receive permissions through membership in the relevant department group. This aligns with least-privilege principles without relying on network-level isolation.

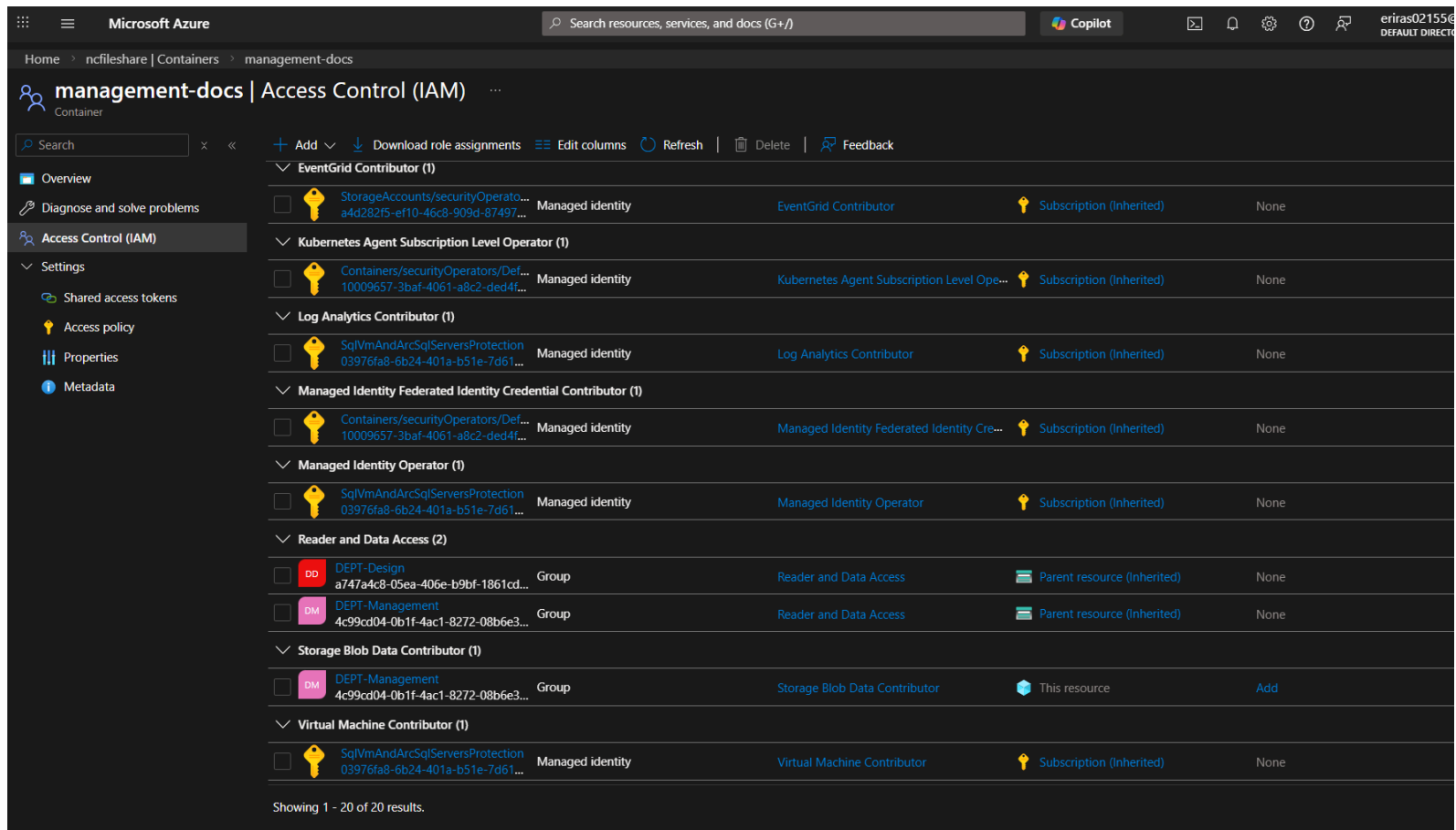
Access control was validated by testing with users from different departments. Design users can reach only the Design-container, Management users can reach only the Management-container, and users outside these groups are unable to access any containers, confirming correct departmental isolation.

Figure 17 RBAC configuration for the Design container.

The screenshot shows the Microsoft Azure portal interface for the 'design-templates' container. The 'Access Control (IAM)' section is selected, displaying a list of role assignments. The roles are categorized into several groups, including EventGrid Contributor, Kubernetes Agent Subscription Level Operator, Log Analytics Contributor, Managed Identity Federated Identity Credential Contributor, Managed Identity Operator, Reader and Data Access (for DEPT-Design and DEPT-Management groups), Storage Blob Data Contributor, and Virtual Machine Contributor. The 'Storage Blob Data Contributor' role is highlighted with a red 'DD' icon. The interface includes a search bar, navigation menu, and a list of role assignments with details such as role name, scope, and permissions.

Role	Scope	Permissions	Assignment
EventGrid Contributor (1)			
StorageAccounts/securityOperators/Def...	Managed identity	EventGrid Contributor	Subscription (Inherited)
Kubernetes Agent Subscription Level Operator (1)			
Containers/securityOperators/Def...	Managed identity	Kubernetes Agent Subscription Level Ope...	Subscription (Inherited)
Log Analytics Contributor (1)			
SqIvMAndArcSqlServersProtection	Managed identity	Log Analytics Contributor	Subscription (Inherited)
Managed Identity Federated Identity Credential Contributor (1)			
Containers/securityOperators/Def...	Managed identity	Managed Identity Federated Identity Cre...	Subscription (Inherited)
Managed Identity Operator (1)			
SqIvMAndArcSqlServersProtection	Managed identity	Managed Identity Operator	Subscription (Inherited)
Reader and Data Access (2)			
DEPT-Design	Group	Reader and Data Access	Parent resource (Inherited)
DEPT-Management	Group	Reader and Data Access	Parent resource (Inherited)
Storage Blob Data Contributor (1)			
DEPT-Design	Group	Storage Blob Data Contributor	This resource
Virtual Machine Contributor (1)			
SqIvMAndArcSqlServersProtection	Managed identity	Virtual Machine Contributor	Subscription (Inherited)

Figure 18 RBAC configuration for the Management container.



Role	Assignment	Role Name	Subscription	Permissions
EventGrid Contributor (1)				
 StorageAccounts/securityOperators/Def...	Managed identity	EventGrid Contributor	Subscription (Inherited)	None
Kubernetes Agent Subscription Level Operator (1)				
 Containers/securityOperators/Def...	Managed identity	Kubernetes Agent Subscription Level Ope...	Subscription (Inherited)	None
Log Analytics Contributor (1)				
 SqlVmAndArcSqlServersProtection	Managed identity	Log Analytics Contributor	Subscription (Inherited)	None
Managed Identity Federated Identity Credential Contributor (1)				
 Containers/securityOperators/Def...	Managed identity	Managed Identity Federated Identity Cre...	Subscription (Inherited)	None
Managed Identity Operator (1)				
 SqlVmAndArcSqlServersProtection	Managed identity	Managed Identity Operator	Subscription (Inherited)	None
Reader and Data Access (2)				
 DEPT-Design	Group	Reader and Data Access	Parent resource (Inherited)	None
 DEPT-Management	Group	Reader and Data Access	Parent resource (Inherited)	None
Storage Blob Data Contributor (1)				
 DEPT-Management	Group	Storage Blob Data Contributor	This resource	Add
Virtual Machine Contributor (1)				
 SqlVmAndArcSqlServersProtection	Managed identity	Virtual Machine Contributor	Subscription (Inherited)	None

The setup was validated through tests showing that each department can only access its own container, while users without permissions see no containers.

Figure 19 Blocked access for users without the correct RBAC-group.

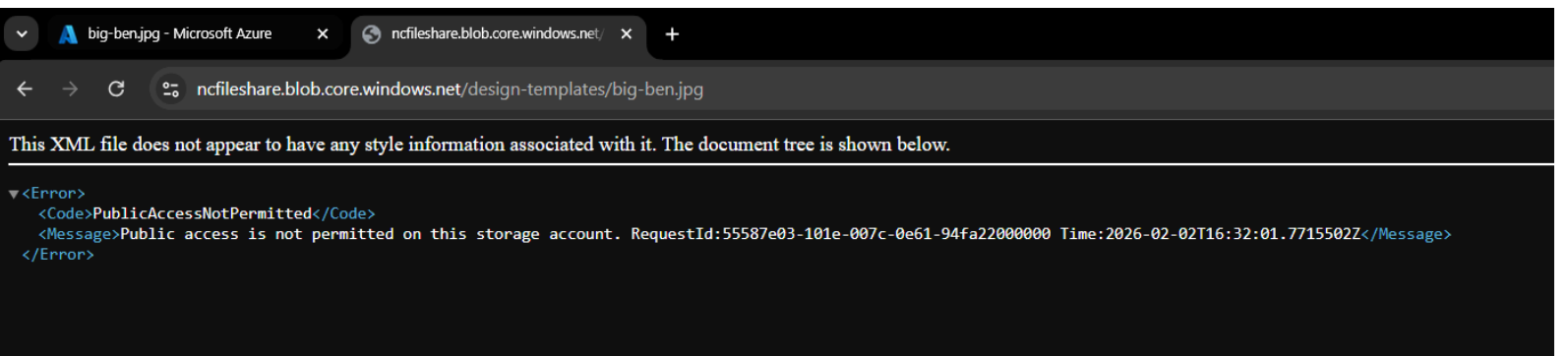


Figure 20 Design user access to Design-container.

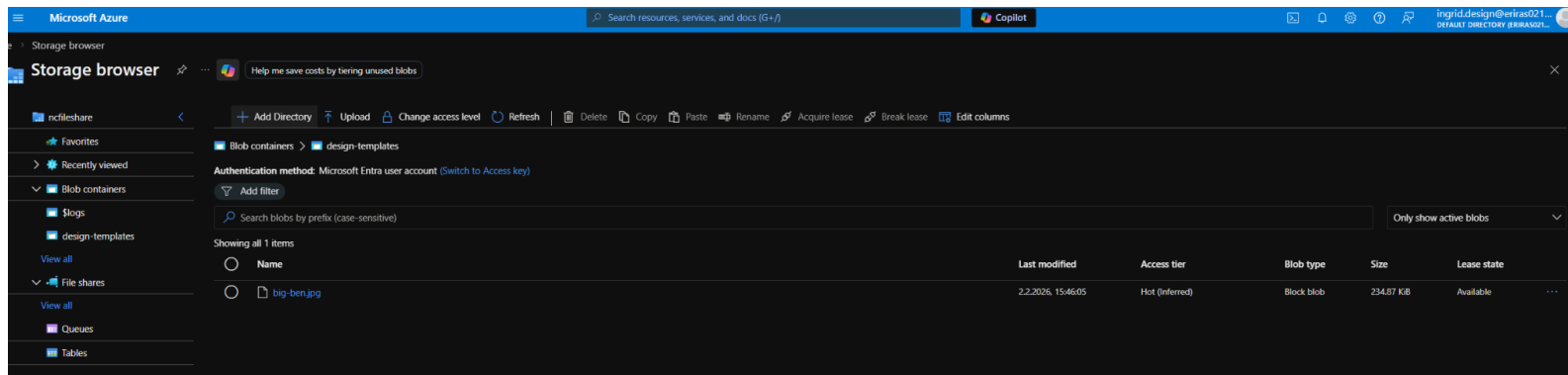


Figure 21 Management user access to Management-container.

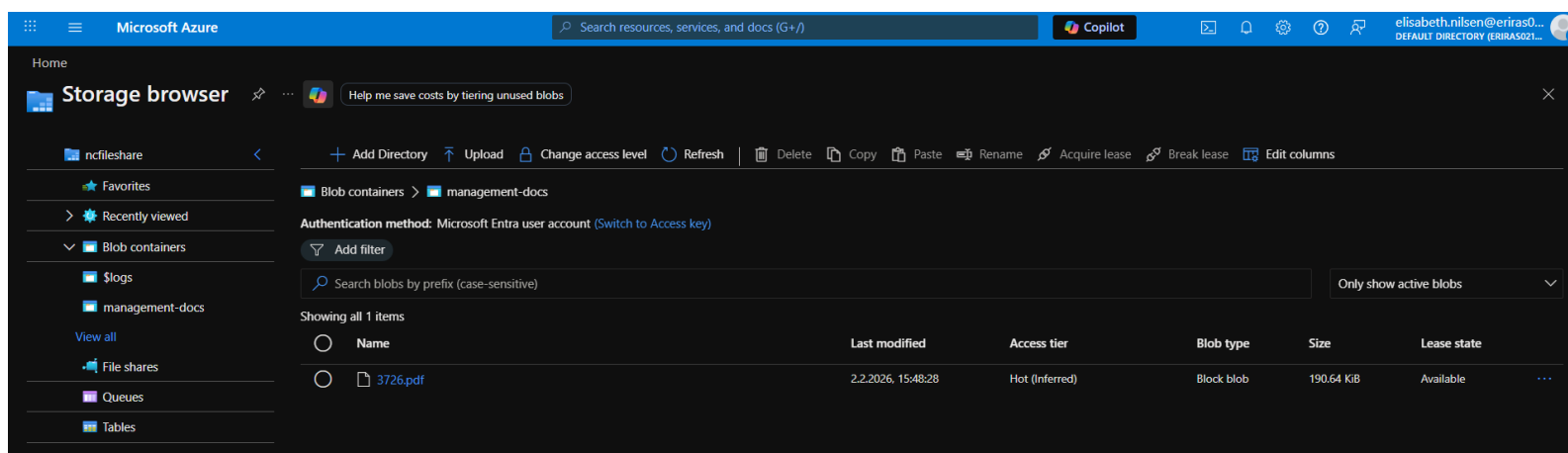
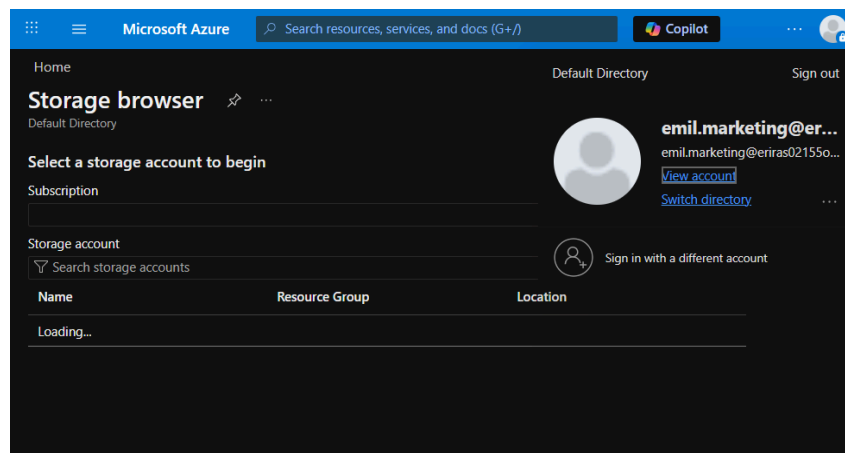


Figure 22 User without correct groups: containers hidden.



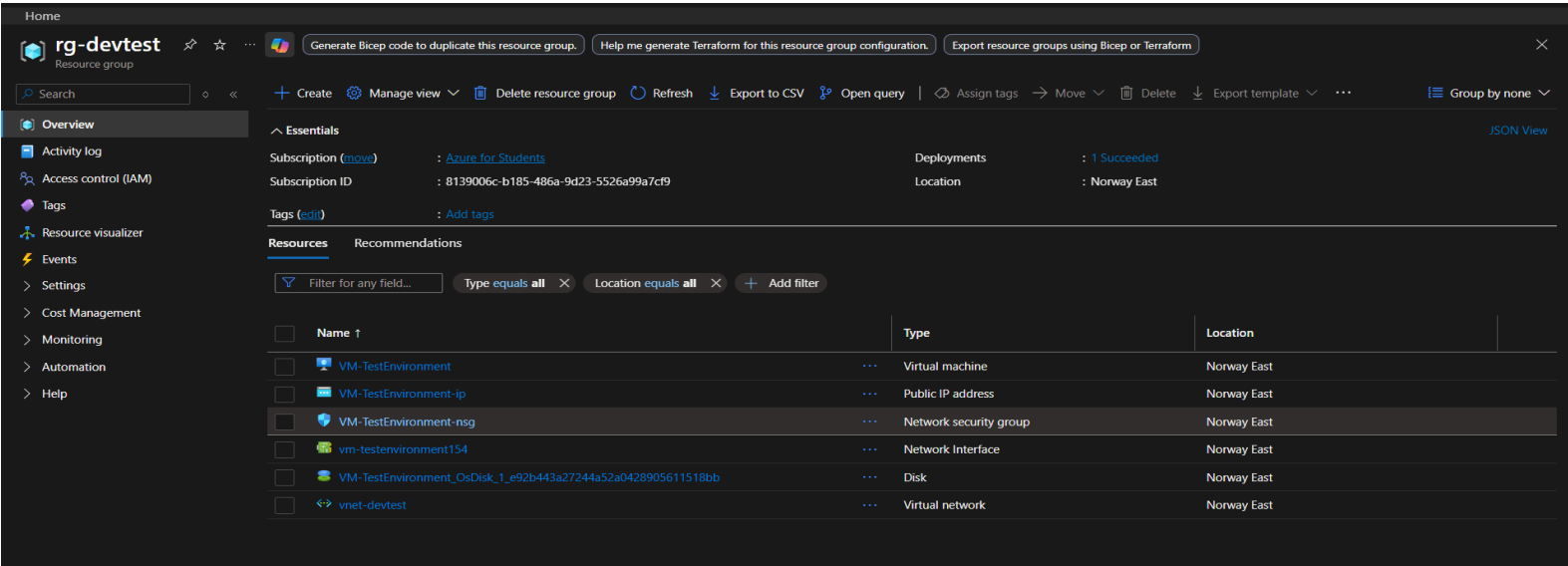
Overall, this approach offers a secure and operationally simple document-sharing solution that meets NerdyClouds' current requirements. More complex alternatives, such as traditional file servers or additional VNet separation, were deliberately excluded to avoid unnecessary complexity and overengineering at this stage.

2.4.3 Isolated Dev/Test Environment

Developers require an isolated environment for development and testing that must not impact production systems. To meet the requirement, a DevTest-environment is deployed using a virtual machine hosted in an isolated VNet with its own resource group.

The isolated environment has no direct connectivity to the Production-VNet. As a result, the design is intended to prevent non-production workloads from impacting production services by removing direct network connectivity (Microsoft, n.d.-b). Figure 23 shows the DevTest-virtual machine (VM) deployed in a resource group, separated from production resources.

Figure 23 DevTest-VM deployed in an isolated resource group.



Access to the VM is controlled using Entra ID and RBAC (Microsoft, 2024a). Developers authenticate using their individual identities in Azure and are granted access through group-based role assignments, avoiding shared credentials and aligning with the identity-first access model. Figure 24 shows the role assignment used to grant developers access to the DevTest-virtual machine.

Figure 24 RBAC access restricted to RBAC-DevTest-Contributors.

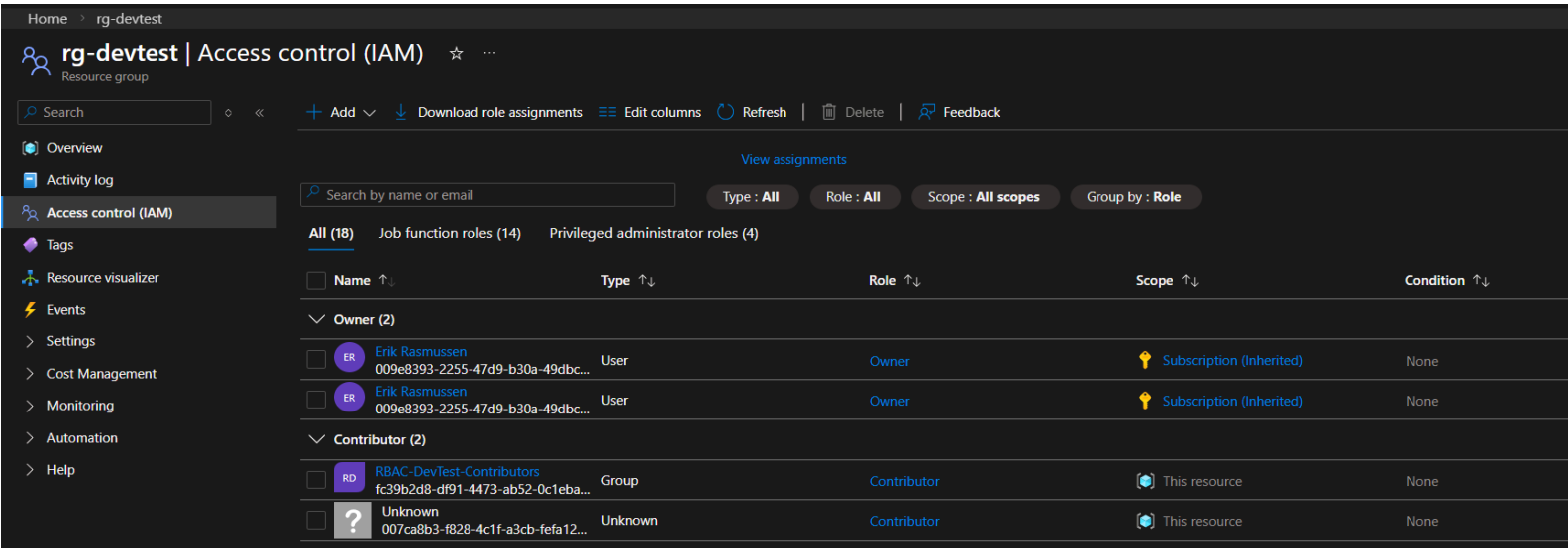
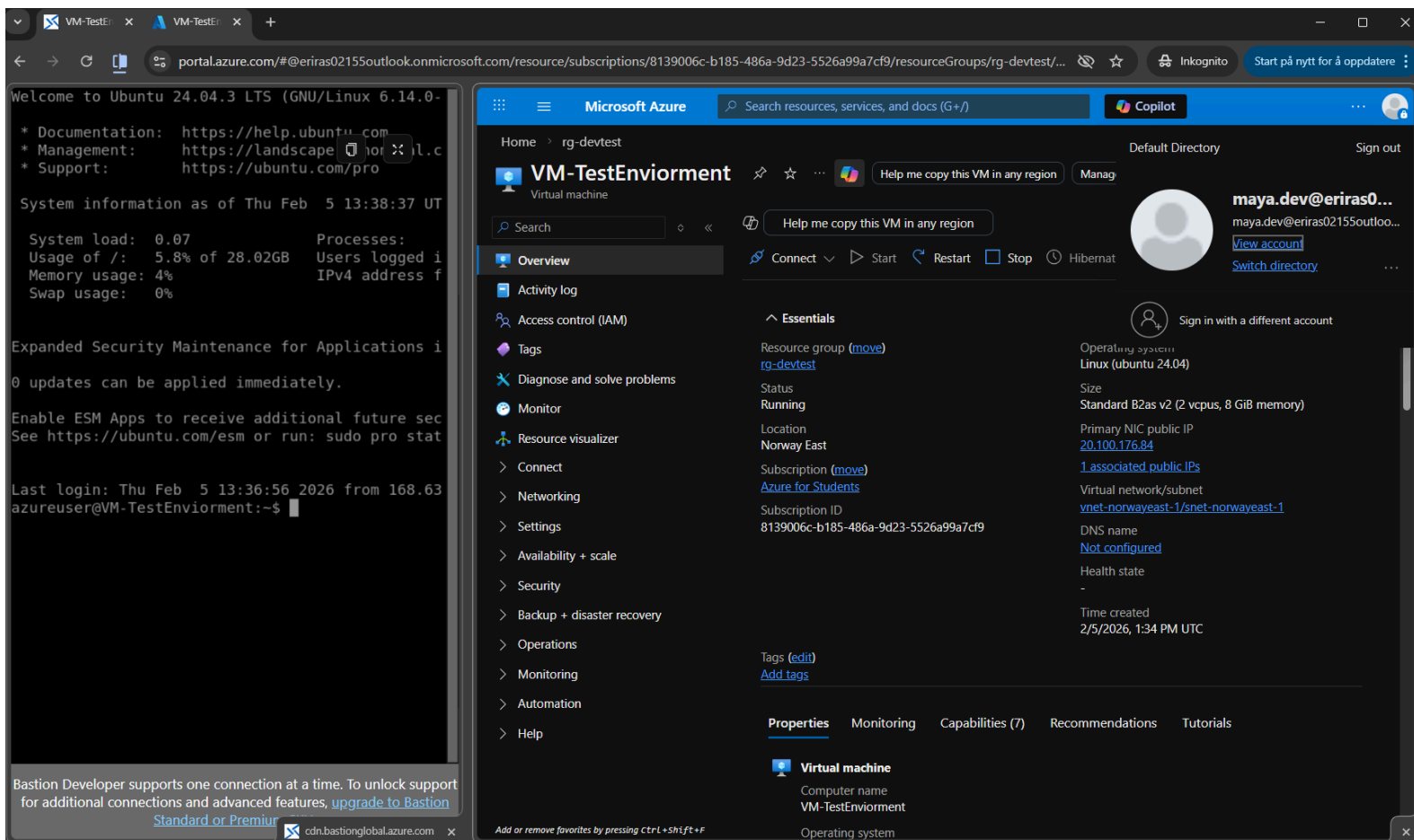


Figure 25 TestEnvironment-VM login validated: RBAC-DevTest-Contributor is able to login (and rg-devtest resource access) as part of the isolated environment.



The DevTest-environment is intentionally designed with lower availability requirements, as prioritising high availability in non-production environments would add unnecessary cost and complexity. Its primary purpose is experimentation and testing rather than business continuity. Cost and complexity are kept under control by using only the resources needed for this environment.

2.4.4 Public Support Chat

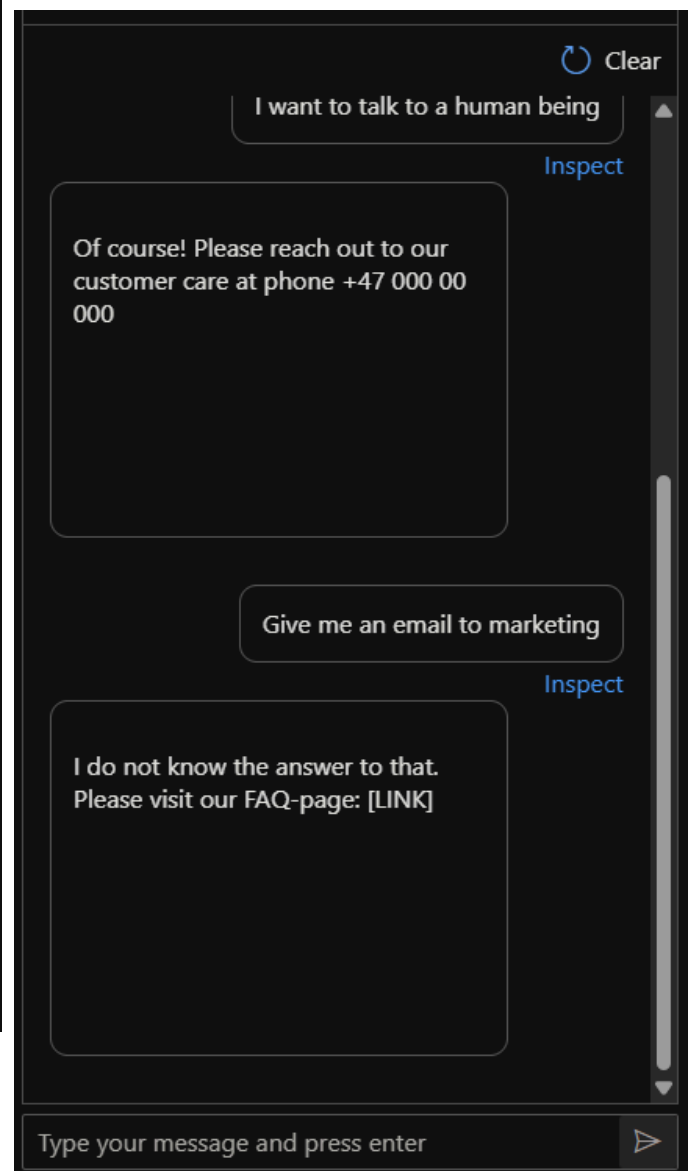
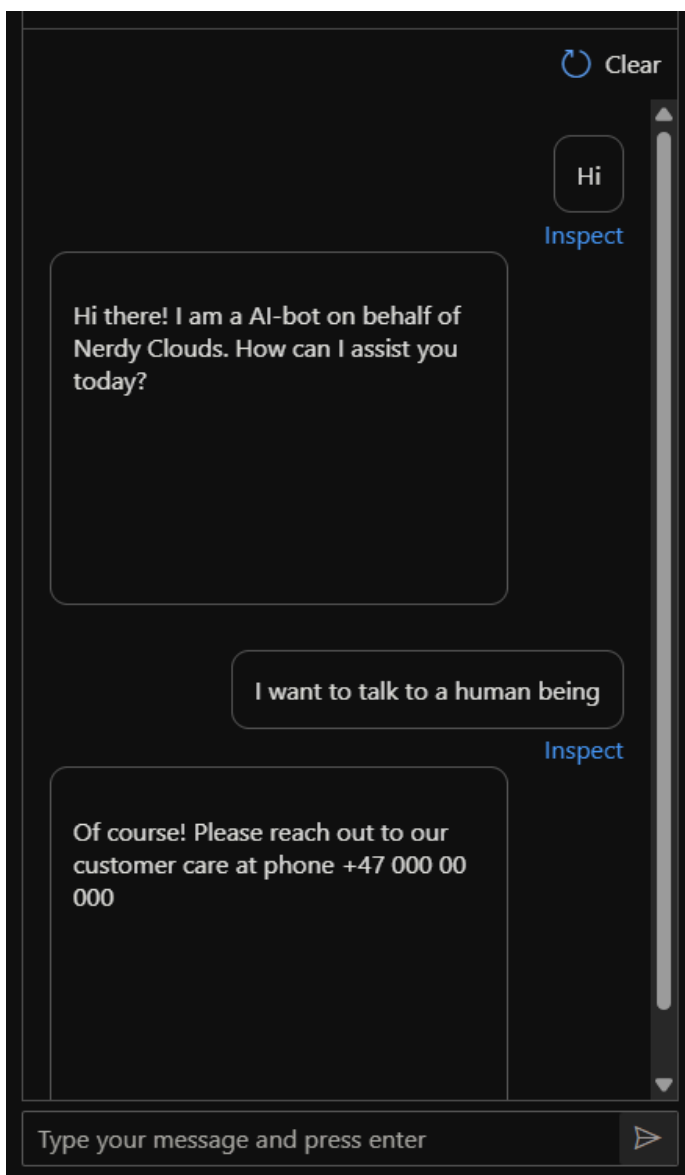
NerdyClouds requires a customer-facing chat-service to handle support-related enquiries. The primary goal of this service is to provide responses to frequently asked questions. It is intended to reduce manual workload for human support staff and aims to improve customer experience through immediate availability. The support chat is operated by the Sales department.

The customer support chat uses the Question Answering capability in Azure AI Language to return verified answers from an approved knowledge base. This service allows natural language questions to be matched against a predefined knowledge base, returning relevant answers even when questions are phrased differently (Microsoft, 2025e).

This chat service is designed to handle predictable and repetitive customer enquiries without requiring custom application development. It is intentionally designed as a public-facing component and does not require user authentication. Security is enforced by limiting the service to a controlled knowledge base only (Microsoft, 2025e). In this configuration, the chat has no access to internal systems, documents, or identities, helping reduce the risk of sensitive data exposure.

This approach provides a simple, scalable customer support capability while aligning with the overall architecture by keeping public services isolated from internal services. Additional AI functionality can be introduced later by extending Azure AI Services without redesigning the underlying solution.

Figures 26 and 27 illustrate responses returned from the deployed knowledge base when queried using natural language input.



2.4.5 Internal AI Assistant

NerdyClouds requires an internal AI assistant to support management with fast access to information, summaries, and decision support without exposing internal discussions or organisational context to external services. The goal is to reduce time spent searching across documentation and to help management produce consistent internal communication, while keeping strict control over who can access the tool.

In the Proof-of-Concept, the assistant is deployed in Microsoft Foundry using a Large Language Model (LLM). Microsoft Foundry is used as the hosting environment for the model and related testing (Microsoft, 2026; Jackson, 2025). The assistant is intended only for internal use for the management-department.

In the Proof-of-Concept, access is scoped through Entra ID group-based RBAC so that only the DEPT-Management group is granted permissions to the Foundry resource and project. Which aligns with least-privilege principles by managing access through group membership rather than individual assignments.

The Proof-of-Concept was validated by confirming the model deployment in Foundry and verifying that access is restricted to the DEPT-Management group through group-based RBAC. Finally, the assistant was validated by signing in as a management user and running a test prompt in the Foundry chat playground.

Figure 28 AI Assistant-Model deployed in Microsoft Foundry.

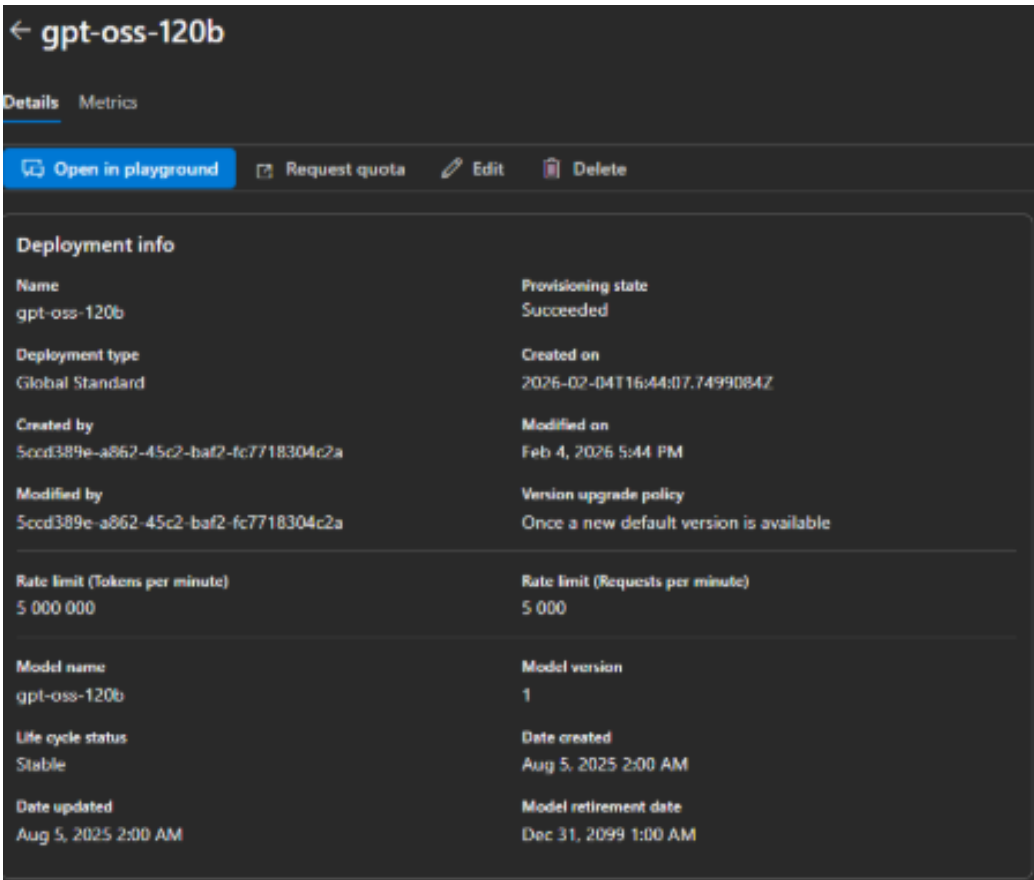
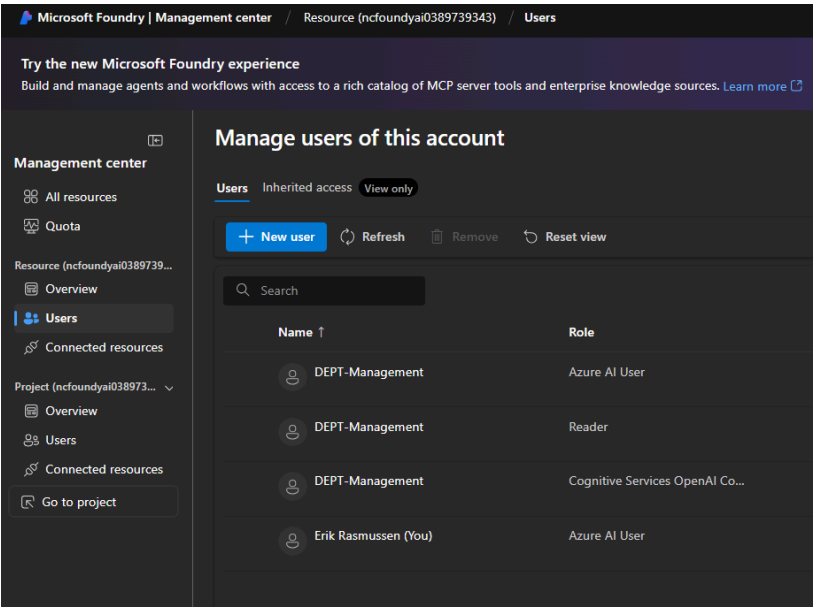
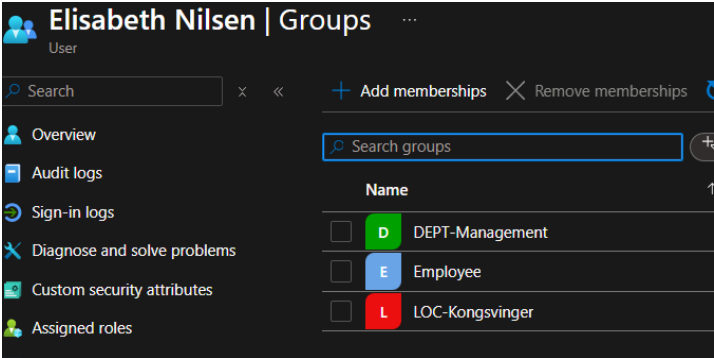
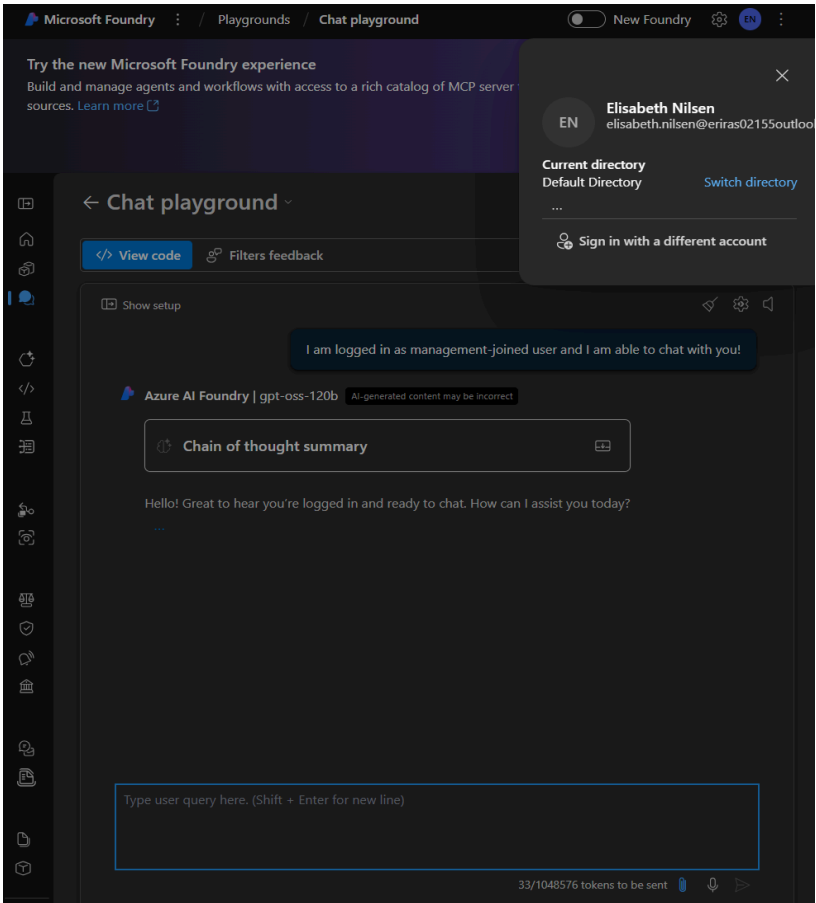


Figure 29 RBAC access restricted to DEPT-Management.



Figures 30 and 31 Management user verified access in Foundry chat playground.



Overall, the department services support secure access to internal applications, document sharing, and support channels without relying on inconsistent, manual processes. For NerdyClouds, this helps reduce time spent on manual access handling and helps departments work faster with clear ownership and separation.

2.5 Data and Analytics

Section 2.5 explains how data is collected, stored, and prepared to support reporting, analytics, and future AI-driven insights across the organisation. In this design, data is treated as a single, trusted source for decision-making, improving visibility across departments.

The focus is on establishing a simple and scalable data foundation using cloud-native services. The design prioritises clear structure, reduced management effort, and alignment with identity-first security principles. At the same time, it keeps the organisation well prepared for future analytics and AI extensions.

2.5.1 Data Lake Architecture (Blob Storage)

A data lake provides a central location for storing data, making organisational data accessible for analytics and advanced analysis processes (Zagan & Danubianu, 2023). Azure Blob Storage is selected as a scalable storage foundation with integration into Azure analytics services.

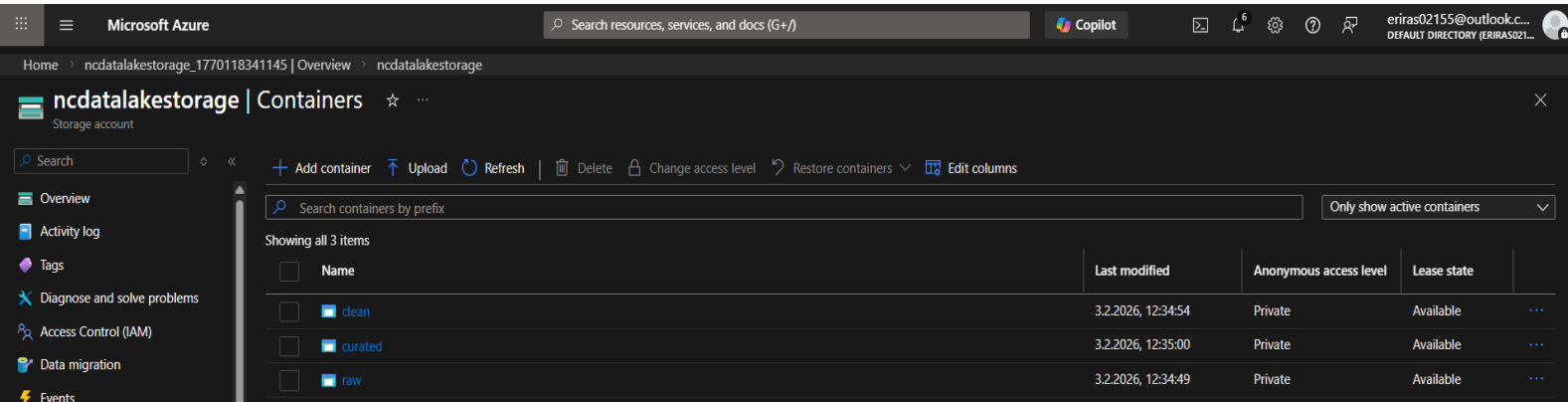
NerdyClouds’ data lake is implemented using Blob Storage and organised into three zones:

- 1. **RAW** original data as ingested
- 2. **CLEAN** standardised and validated data
- 3. **CURATED** refined data ready for reporting and analytics

The data structure is designed to support a clear data lifecycle and makes it easier to transition from raw files to analytics-ready datasets.

By keeping all analytical data in Blob Storage, the organisation can consolidate data from multiple sources into a single, controlled data foundation. The data foundation is intended to support reporting and analytics in tools such as Power BI and Microsoft Fabric (Fabric), while remaining flexible for future AI and automation use cases (Microsoft, n.d.-d; Microsoft, 2025m). Power BI and Fabric are covered further in Section 2.5.2

Figure 32 Data lake zones: **RAW / CLEAN / CURATED**.



2.5.2 Analytics with Fabric

Fabric is used as a Proof-of-Concept to validate how data stored in the data lake can be analysed and visualised. The goal of this section is not to demonstrate a full analytics platform, but to confirm that the data foundation established in Section 2.5.1 can be used to generate insights.

In this Proof-of-Concept, sample data was ingested into a Fabric Lakehouse, where it can be queried using SQL and explored through built-in analytics capabilities (Microsoft, 2025m). This aims to be a repeatable way to analyse data without duplicating storage or introducing additional infrastructure.

Figure 33 Dataset ingested into a Fabric Lakehouse for analysis.

Explorer

taxi-rides

SQL query 1

+ Warehouses

Test

Schemas

dbo

Tables

taxi-rides

Views

Functions

Stored Proce...

INFORMATION_...

queryinsights

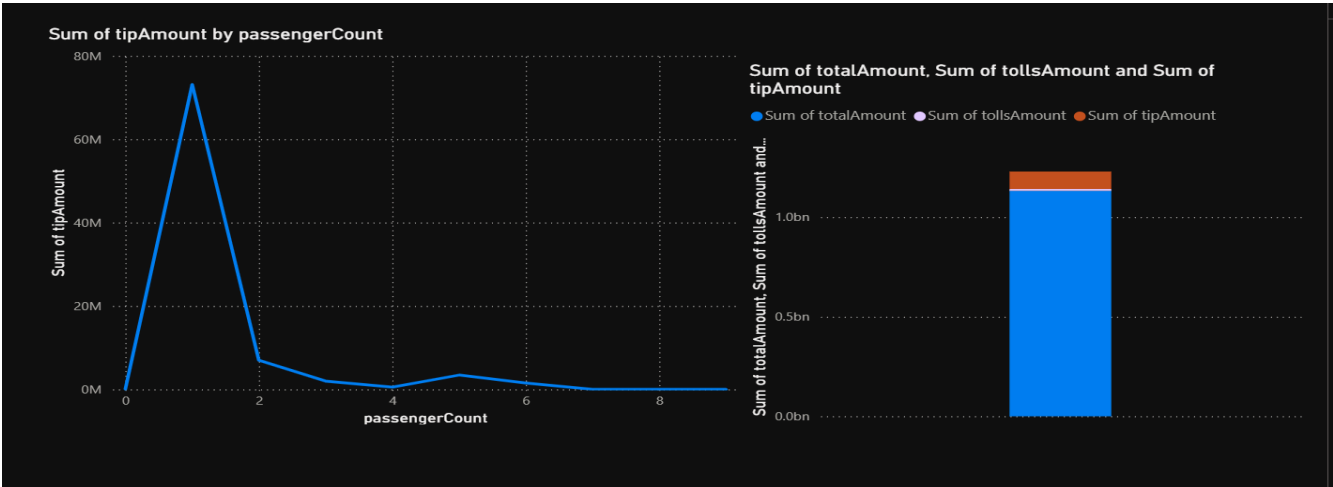
Data preview - taxi-rides

Showing 1000 rows

Search

	123	vendorID	123	lpepPick...	123	lpepDrop...	123	passenge...	12F	tripDista...	ABC	puLocati...	ABC	doLocati...	12F	pickupLo...	12F	pickupLa...	12F	dropoffL...	12F	dropoffL...	123	rateCode...	ABC	storeAnd...
1	2		2017-08-02 1...	2017-08-02 1...	5		0.93	42		42		NULL	NULL	NULL	NULL	1		N								
2	2		2017-08-02 1...	2017-08-02 1...	5		0.65	75		74		NULL	NULL	NULL	NULL	1		N								
3	2		2017-08-02 1...	2017-08-02 1...	1		1.38	255		112		NULL	NULL	NULL	NULL	1		N								
4	2		2017-08-02 1...	2017-08-02 1...	1		3.85	112		100		NULL	NULL	NULL	NULL	1		N								
5	2		2017-08-02 1...	2017-08-02 1...	1		3.08	17		33		NULL	NULL	NULL	NULL	1		N								
6	2		2017-08-02 1...	2017-08-02 1...	1		1.3	33		40		NULL	NULL	NULL	NULL	1		N								
7	2		2017-08-02 1...	2017-08-02 1...	1		2.75	42		236		NULL	NULL	NULL	NULL	1		N								
8	2		2017-08-02 1...	2017-08-02 1...	3		0.92	97		49		NULL	NULL	NULL	NULL	1		N								
9	2		2017-08-02 1...	2017-08-02 1...	3		0.88	97		66		NULL	NULL	NULL	NULL	1		N								
10	2		2017-08-02 1...	2017-08-02 1...	3		1.99	66		40		NULL	NULL	NULL	NULL	1		N								
11	1		2017-08-02 1...	2017-08-02 2...	1		1	69		42		NULL	NULL	NULL	NULL	1		N								
12	1		2017-08-02 1...	2017-08-02 1...	2		3.5	82		192		NULL	NULL	NULL	NULL	1		N								
13	2		2017-08-02 1...	2017-08-02 1...	3		2.13	97		232		NULL	NULL	NULL	NULL	1		N								

Figure 34 Example Power BI visualisation created from data stored in the Fabric Lakehouse.

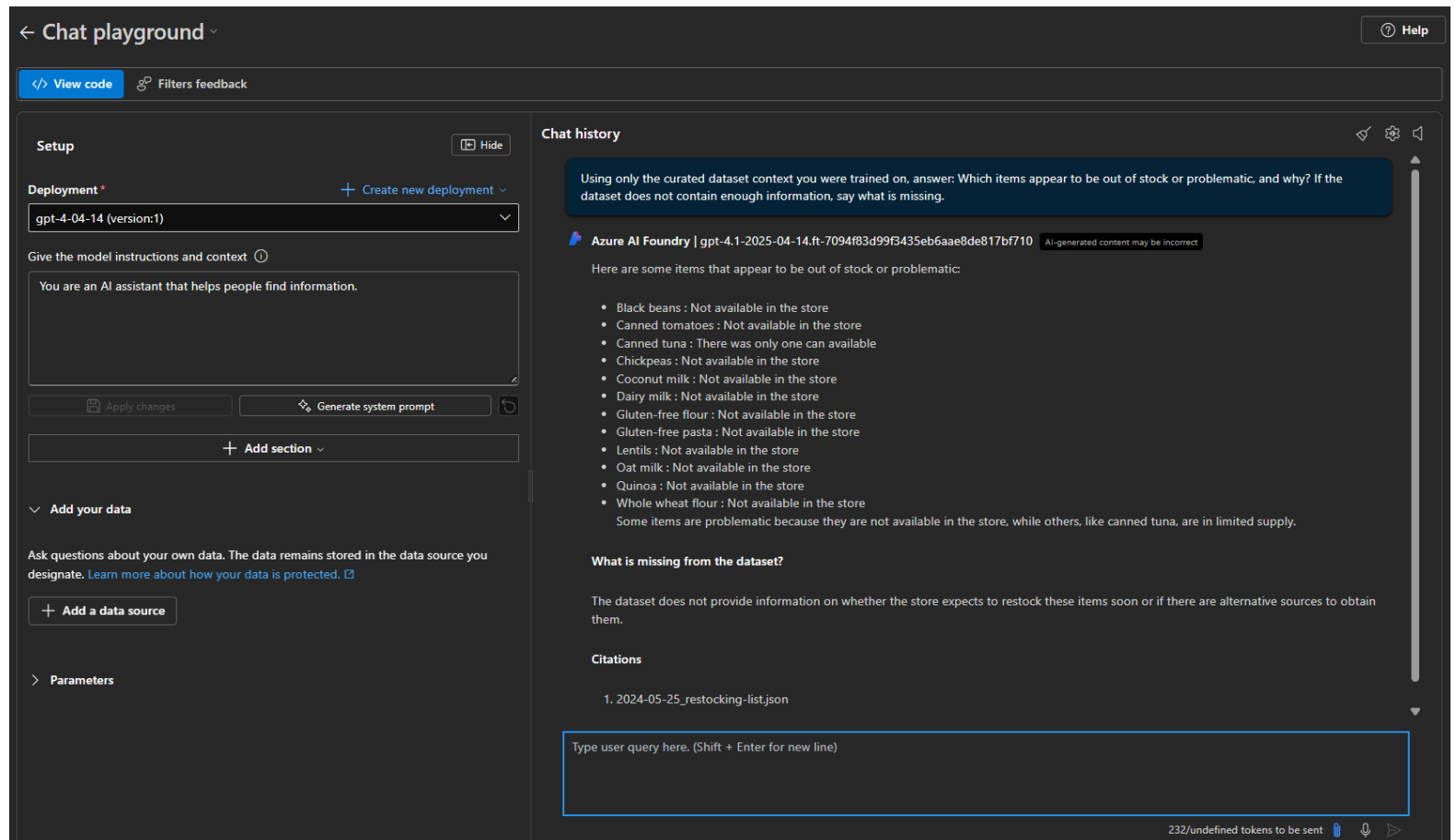


The results are visualised using Power BI to show how insights can be communicated effectively to stakeholders (Microsoft, n.d.-d). For NerdyClouds, the same workflow could be applied to operational data such as support tickets, usage patterns, or service performance metrics, supporting planning and decision-making without manual reporting.

2.5.3 AI and Data

As NerdyClouds stores approved datasets in the CURATED zone, the organisation establishes a foundation for LLM-based analysis. In this Proof-of-Concept, the model was used to generate summaries, identify unusual values or trends, and answer questions in natural language. In this demonstration, a sample dataset was used to validate model deployment and prompting, as internal curated data is limited.

Figure 35 LLM deployment and prompt validation using curated sample data.



If this capability proves valuable, the next phase can expand on curated blob-based data by introducing services such as Azure AI Search to improve discovery and support broader, permission-aware exploration (Microsoft, 2025n).

Overall, the data and AI components are designed to provide a controlled path from curated data to dashboards and guided information access for management and support use cases.

2.6 Use of AI

AI was used as a support-tool to assist with structuring, reviewing, and improving the clarity of the written report. It was not used to perform technical implementations or configure services in Azure. All practical work was completed manually by the author.

The tool was primarily used to help organise the report so that all required topics were covered in the correct order and aligned with the assignment requirements. Suggestions related to structure and wording were reviewed and adjusted by the author before being applied.

In addition, the tool was used for language support, such as correcting grammar, improving sentence clarity, and identifying any unclear formulations. It was also used to assist with formatting references according to APA-7 standards based on source information provided by the author. The tool was not used to select sources, write technical explanations, or carry out hands-on work. All sources were identified by the author, and all proof-of-work, testing, screenshots, and validation steps were performed directly in the Azure environment.

All suggestions were critically reviewed and verified. Any suggestions that did not match the implemented solutions or intended meaning were modified or removed. The main learning outcome from this process was understanding how such tools can be used as controlled writing support, while maintaining full responsibility for technical decisions and final content.

2.7 Cloud Journey Reflection

The cloud journey has shifted NerdyClouds from fragmented, infrastructure-driven operations to a more structured and identity-centric way of working. Access control, security, and availability are consistently enforced across services, reducing operational risk and improving predictability. This has also lowered day-to-day administrative tasks and increased confidence in the environment.

The current environment is stable, functional, and aligned with the company's requirements. Core services are operational, security controls are in place, and the architecture is designed with scalability and future growth in mind. The solution meets availability targets and supports both on-site and remote work scenarios.

A deliberate hybrid approach has been implemented to balance control, risk, and flexibility. The environment is built around a hub-and-spoke network model with clear separation of workloads, identity-first security, and managed cloud services where they reduce complexity. Data foundations have been established to support future analytics and insights.

The logical next step is a gradual reduction of on-premises dependencies as hardware reaches end of life. This enables a controlled transition toward a predominantly cloud-based environment, driven by business timing rather than forced migration. Future work should focus on expanding data usage, automation, and insight-driven services on top of the existing foundation.

3 Summary and Conclusions

This report presented a cloud and hybrid target architecture for NerdyClouds with a focus on minimal ongoing administration, secure access control, reliable service availability, and monitoring. Azure services are used where they reduce manual work and simplify day-to-day operations, supported by a controlled identity and connectivity foundation.

The solution meets the project requirements through a consistent design across networking, identity and access management, department services, and data and analytics. The foundation combines a clear virtual network and DNS design with an identity-first model using Entra ID, groups, and RBAC, supporting controlled access for both internal and remote users.

Department-facing services include an internal web application, secure document sharing using Azure Storage and Blob containers, and an isolated DevTest environment. A blob-based data foundation and analytics insights using Fabric and Power BI demonstrate a practical path from curated data to dashboards. Two AI-enabled services are also included: a public support chat using Azure AI Language (Question Answering) and a restricted internal AI assistant hosted in Microsoft Foundry. The department services, analytics, and AI components demonstrate how a single identity and network foundation supports daily work across teams without unnecessary complexity.

Proof-of-Concept validation is included to demonstrate that access control and service behaviour work as intended. The validation focused on authentication, authorisation, and department separation.

Table 1. Key Decisions

Table 1 summarises key design choices, who they affect, and what is lost without them.

Service	Who it impacts	Benefit and change	What you lose if not implemented
Azure Virtual Network (hub-and-spoke)	IT operations, all departments	Centralises connectivity and security control, with clear separation between Production and Dev/Test.	Network sprawl and weaker separation, making change and troubleshooting slower.
Azure VPN Gateway (site-to-site)	On-prem users, IT operations	Secure hybrid connectivity so cloud services can be used without changing daily work patterns.	Cloud services become isolated, increasing workarounds and reducing hybrid value.
Azure Private DNS + Private DNS Zones	Support team, app owners	Consistent internal name resolution across VNets, improving service stability and reducing incidents.	More name-resolution errors and slower recovery during incidents.
Microsoft Entra ID + Entra Connect	All users, IT security	Single sign-in and central identity control while keeping on-prem lifecycle as the source of truth.	Fragmented accounts, weaker governance, and more manual identity work.
Azure RBAC with Entra security groups	IT operations, auditors, managers	Access is tied to role groups, improving least-privilege enforcement and auditability.	Permission sprawl and harder access reviews, increasing over-access risk.
Azure App Service (internal web app) with Entra auth	Management users, IT operations	Managed hosting and integrated authentication, reducing server maintenance and speeding delivery.	More server administration effort and higher risk of patching gaps.
Azure Storage Account (Blob) with containers + ZRS	Design and Management departments, IT operations	Department isolation through container-scoped RBAC and resilient storage for critical documents.	Manual sharing, unclear access boundaries, and weaker availability for shared content.
Microsoft Fabric + Power BI (analytics PoC)	Management, analysts, department leads	Validates a repeatable path from curated data to dashboards for decision support.	Slower reporting cycles and more manual spreadsheet-based analysis.

Some components are described as target design and are not fully implemented in the Proof-of-Concept environment. This includes Conditional Access enforcement, VPN-based hybrid connectivity (S2S and P2S), and related hybrid DNS extensions such as conditional forwarding. In addition, internal DNS hostnames for App Service are described conceptually rather than implemented. Future improvements can build on the established data foundation, including more advanced analytics and controlled internal search capabilities.

Self-Reflection

This project changed how I prioritise architecture work. I expected service selection to be the hardest part. In practice, the hardest part was defining the foundation: identity groups, RBAC scope, and network segmentation. When these were clear, later decisions were easier to implement and validate.

Zero Trust became a practical concept. Least privilege only worked consistently when access was granted through groups and enforced with RBAC at the right scope. The work made it clear that shortcuts in access assignment quickly lead to unclear boundaries and exceptions that are hard to maintain.

The work also showed that there is rarely a single correct answer. Most choices involved trade-offs between security, simplicity, cost, and availability. The target was therefore a design that is predictable to operate and easy to manage for a small organisation.

Next time, I would start smaller and prove the basics first. If identity flow, DNS resolution, and routing behave as expected early on, the rest of the design becomes easier to build on and easier to trust.

4 References

- Artac, M., Borovsak, T., Di Nitto, E., Guerriero, M., & Tamburri, D. A. (2017). DevOps: Introducing infrastructure-as-code. In *Proceedings of the 2017 IEEE/ACM 39th International Conference on Software Engineering Companion (ICSE-C)* (pp. 497–498). IEEE.
https://www.researchgate.net/profile/Damian-Tamburri/publication/318124847_DevOps_Introducing_Infrastructure-as-Code/links/5b17c6bbaca272021ce9147e/DevOps-Introducing-Infrastructure-as-Code.pdf
- Beeram, S. (2025). Zero trust identity management with Azure Entra and Conditional Access. *International Journal of Emerging Trends in Computer Science and Information Technology*, 6(4), 173–175. <https://www.ijetcsit.org/index.php/ijetcsit/article/download/532/478>
- Borra, P. (2024). Microsoft Azure networking: Empowering cloud connectivity and security. *International Journal of Advanced Research in Science, Communication and Technology*, 4(3), 469–475. <https://papers.ssrn.com/sol3/Delivery.cfm?abstractid=4914155>
- Howard, M., Gantenbein, H., & Curzi, S. (2023). *Designing and developing secure Azure solutions*. Pearson Education.
https://ptgmedia.pearsoncmg.com/images/9780137908752/samplepages/9780137908752_Sample.pdf
- Immidiseti, V. R. (2021). Azure private DNS zones: Architecture, implementation best practices, and organizational use cases. *International Journal of Multidisciplinary Research and Growth Evaluation*, 2(6), 446–449.
https://www.allmultidisciplinaryjournal.com/uploads/archives/20250319172048_F-21-109.1.pdf
- Jackson, F. (2025). A unified governance architecture for agentic AI: Integrating Microsoft Azure AI Foundry, cloud-native guardrails, frontier AI safety research, zero-trust principles, and the 4th.GRC governance engine (Working paper). Fourth Industrial Systems Corporation.
https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5819502
- Lane, M., Shrestha, A., & Ali, O. (2017). Managing the risks of data security and privacy in the cloud: A shared responsibility between the cloud service provider and the client organisation [Conference paper]. The Bright Internet Global Summit 2017, Seoul, South Korea.
<https://research.usq.edu.au/download/e62bddbebbcae2d7e07584aabdd79d94a1339a1016e116eb1a2ad97f457550df/124402/B5.%20Managing%20the%20Risks%20of%20Data%20Security%20and%20Privacy%20in%20the%20Cloud.pdf>
- Microsoft. (n.d.-a). *Describe the benefits of security and governance in the cloud*. Microsoft Learn.
<https://learn.microsoft.com/en-us/training/modules/describe-benefits-use-cloud-services/4-security-governance-cloud>

Microsoft. (n.d.-b). Hub-spoke network topology in Azure. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/architecture/networking/architecture/hub-spoke>

Microsoft. (n.d.-c). Design a hybrid domain name system solution with Azure. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/architecture/hybrid/hybrid-dns-infra>

Microsoft. (n.d.-d). Power BI – Data visualization. Microsoft.
<https://www.microsoft.com/en-us/power-platform/products/power-bi>

Microsoft. (2024a). What is Azure role-based access control (Azure RBAC)? Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/role-based-access-control/overview>

Microsoft. (2024b). Authorize access to blobs using Microsoft Entra ID. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/storage/blobs/authorize-access-azure-active-directory>

Microsoft. (2025a). Subscription deployments with ARM templates. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/azure-resource-manager/templates/deploy-to-subscription?tabs=azure-cli>

Microsoft. (2025b). The principle of least privilege with Microsoft Entra ID governance. Microsoft Learn.
<https://learn.microsoft.com/en-us/entra/id-governance/scenarios/least-privileged>

Microsoft. (2025c). What is Azure Private DNS? Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/dns/private-dns-overview>

Microsoft. (2025d). Azure network security groups overview. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/virtual-network/network-security-groups-overview>

Microsoft. (2025e). What is custom question answering? Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/ai-services/language-service/question-answering/overview>

Microsoft. (2025f). Microsoft Entra Connect Sync: Understand and customize synchronization. Microsoft Learn.
<https://learn.microsoft.com/en-us/entra/identity/hybrid/connect/how-to-connect-sync-what-is>

Microsoft. (2025g). What is Microsoft Entra? Microsoft Learn.
<https://learn.microsoft.com/en-us/entra/fundamentals/what-is-entra>

Microsoft. (2025h). Conditional Access overview. Microsoft Learn.
<https://learn.microsoft.com/en-us/entra/identity/conditional-access/overview>

Microsoft. (2025i). Azure Storage redundancy. Microsoft Learn.
<https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy?toc=%2Fazure%2Fstorage%2Fblobs%2Ftoc.json>

Microsoft. (2025j). App Service overview. Microsoft Learn.

<https://learn.microsoft.com/en-us/azure/app-service/overview>

Microsoft. (2025k). Authentication and authorization in Azure App Service and Azure Functions. Microsoft Learn.

<https://learn.microsoft.com/en-us/azure/app-service/overview-authentication-authorization>

Microsoft. (2025l). Azure Monitor overview. Microsoft Learn.

<https://learn.microsoft.com/en-us/azure/azure-monitor/fundamentals/overview>

Microsoft. (2025m). What is a lakehouse in Microsoft Fabric? Microsoft Learn.

<https://learn.microsoft.com/en-us/fabric/data-engineering/lakehouse-overview>

Microsoft. (2025n). What is Azure AI Search? Azure Docs.

<https://docs.azure.cn/en-us/search/search-what-is-azure-search?tabs=indexing%2Cquickstarts>

Microsoft. (2025o). About VPN Gateway. Microsoft Learn.

<https://learn.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-about-vpngateways>

Microsoft. (2026). What is Microsoft Foundry? Microsoft Learn.

<https://learn.microsoft.com/en-us/azure/ai-foundry/what-is-foundry?view=foundry-classic>

Mumbere, P., Obira, O., & Ndyabahika, P. (2024). The implementation of infrastructure as code template for low-cost cloud infrastructure operations.

https://www.researchgate.net/profile/Okello-Obira/publication/387268393_The_Implementation_of_Infrastructure_as_Code_Template_for_Low-cost_Cloud_Infrastructure_Operations/links/679deb0052b58d39f26139ed/The-Implementation-of-Infrastructure-as-Code-Template-for-Low-cost-Cloud-Infrastructure-Operations.pdf

National Institute of Standards and Technology [NIST]. (2011). The NIST definition of cloud computing (NIST Special Publication 800-145). <https://doi.org/10.6028/NIST.SP.800-145>

National Institute of Standards and Technology [NIST]. (2020a). Security and privacy controls for information systems and organizations (NIST Special Publication 800-53, Revision 5).

<https://doi.org/10.6028/NIST.SP.800-53r5>

National Institute of Standards and Technology [NIST]. (2020b). Zero trust architecture (NIST Special Publication 800-207). <https://doi.org/10.6028/NIST.SP.800-207>

Salko, S. (2024). Building an information secure cloud environment in Azure with Azure Bicep: Case ECB product from Evitec (Master's thesis, Tampere University). Tampere University.

<https://trepo.tuni.fi/bitstream/handle/10024/154779/SalkoSamu.pdf?sequence=2>

Sandhu, R. S. (1997, September 17). Role-based access control. George Mason University.

<https://profsandhu.com/articles/advcom/a98rbac.pdf>

Zagan, E., & Danubianu, M. (2023). Data lake architecture for storing and transforming web server access log files. *IEEE Access*, 11, 40916–40929.

<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10107911>